

PROTECTIVE ORDER MATERIAL

Change 175294 on 2004/06/24 by mmantor@FL_mmantorLT_r400_win
< initial checkin of osm random generator with minor changes to supporting files>

Change 172110 on 2004/06/07 by llefebvr@llefebvr_r400_linux_marlboro
Fixing bad resource hang on the vertex shader.

Change 171407 on 2004/06/02 by llefebvr@llefebvr_r400_linux_marlboro
Fixing Qnan issue

Change 168815 on 2004/05/19 by llefebvr@llefebvr_r400_linux_marlboro
Integrate of the number of RS stations fixes from Xenos.

Change 168560 on 2004/05/18 by llefebvr@llefebvr_r400_linux_marlboro
Back integration of the AddrValid fix from Xenos.

Change 168033 on 2004/05/14 by llefebvr@llefebvr_r400_linux_marlboro
Integration of Randy's Kill maks changes from Xenos.

Change 167975 on 2004/05/14 by llefebvr@llefebvr_r400_linux_marlboro
Blanket integrate of the Xenos emulator files for SQ,SP,SX to R400.

Change 167630 on 2004/05/13 by mmantor@FL_mmantorLT_r400_win
<made alterations to tp_formatter to work in sp standalone environment made vsp more hardware like in pred, kill, mova, write enables and update on testbench>

Change 155924 on 2004/03/17 by llefebvr@llefebvr_r400_emu_montreal
Integration from Xenos of the valid bit change. tb_sqspsx tracker data only, no functionnal change here.

Change 155819 on 2004/03/17 by llefebvr@llefebvr_r400_emu_montreal
Integration of the gpr clamping base address change from Xenos.

Change 154760 on 2004/03/12 by kryan@kryan_r400_win_marlboro_XP
Integrate changes from Xenos/devel to my r400/branch to r400/devel

Integrate Xenos/devel to r400 branch

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- Added YIELD* keywords to the Shader Assembler.
- Added processing for serial bit in SQ and Shader Assembler.
- Changed handling of Control flow constants in Emulator
to more closely match the hardware.

Change 154141 on 2004/03/10 by llefebvr@llefebvr_r400_linux_marlboro
integration of the valid bit bug fix from Xenos

Change 153494 on 2004/03/08 by llefebvr@llefebvr_r400_emu_montreal
Integration of the VS_DONE_EVENT change from Xenos.

Change 152749 on 2004/03/04 by llefebvr@llefebvre_laptop_r400_emu
Integration of the PRED_COND_CALL change to R400.

Change 152663 on 2004/03/04 by llefebvr@llefebvre_laptop_r400_emu
Fixing COND_JMP and COND_CALL predicated. Also fixing bit mask problem in the SX that controls the tracker.

Change 151381 on 2004/02/26 by llefebvr@llefebvre_laptop_r400_emu
Integration of the cond_pred_execute fix from Xenos

Change 151315 on 2004/02/26 by llefebvr@llefebvr_r400_linux_marlboro
Removing the storage for RT parameters in the emulor from the register space and the SX.

Change 151063 on 2004/02/25 by llefebvr@llefebvr_r400_linux_marlboro
Incorporating Tom's changes to the emulator.

Change 149973 on 2004/02/19 by llefebvr@llefebvr_r400_linux_marlboro
Validated Tom's fix.

Change 149972 on 2004/02/19 by llefebvr@llefebvr_r400_emu_montreal
Fixing the control flow machine in the emulator to remove an extra trip to the RS on a COND_PRED_EXEC opcode.

Change 149638 on 2004/02/17 by llefebvr@llefebvr_r400_emu_montreal

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This makes the emulator write to PV/PS on all units even when the operation is predicated. Only the writes to the registers are blocked on predication (and pred vector/kill vector).

Change 149424 on 2004/02/16 by llefebvr@llefebvr_r400_emu_montreal
Fixing REP bug.

Change 149402 on 2004/02/16 by llefebvr@llefebvr_r400_emu_montreal
This fixes the COND_EXEC_PRED opcode so it returns the thread with the correct resource when the predicate vector is dirty.

Change 148881 on 2004/02/12 by llefebvr@llefebvr_r400_emu_montreal
The predicate clean bit was overwritten instead of being "anded" in creating less clause boundaries than it should have on the emulator.

Change 148666 on 2004/02/11 by llefebvr@llefebvr_r400_emu_montreal
This is a fix for the control flow instruction mismatch in the case of a COND_EXEC_PRED opcode. The emulator wasn't updating the resource field while the HW was.

Change 148344 on 2004/02/10 by llefebvr@llefebvr_r400_emu_montreal
Enabling PRED_CLEAN instructions in primlib. Fixed a problem with COND_EXEC_PRED_CLEAN_END opcode in the emulator.

Change 148246 on 2004/02/09 by donaldl@donaldl_xenos_linux_orl
Added flat_shading signal for use in the SX parameter subtract function. (ie. if flat_shading is true, ignore infinity checks; just do subtract. Result should be zero.)

Change 148084 on 2004/02/09 by llefebvr@llefebvre_laptop_r400_emu
Implemented the REP function in the emulator.

Change 147964 on 2004/02/07 by donaldl@donaldl_xenos_linux_orl
Fixed bug when calling parameter_sub during creation of vectors (ie. lessthan0 and greaterpoint5 parameters were swapped).

Change 147952 on 2004/02/07 by mmantor@mmantor_xenos_linux_orl
<added test bench for DiffEng param_sub in the sx for numerical verification>

Change 147697 on 2004/02/06 by llefebvr@llefebvre_laptop_r400_emu

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The emulator now ignores the MSB of the count field.

Change 147181 on 2004/02/04 by llefebvr@llefebvre_laptop_r400_emu
Adding Sq->SX event dump file.

Change 146961 on 2004/02/03 by mearl@mearl_xenos_linux_orl
Updated the emulator to match the hardware.

Change 146954 on 2004/02/03 by llefebvr@llefebvre_laptop_r400_emu
Fixing alloc with no serial folowed by serialized instruction bug (emulator wasn't stopping).

Change 146738 on 2004/02/02 by mmantor@FL_mmantorLT_r400_win
<added checking of vector and scalar results all the time, constant address and export control&fog overlay, More values are randomized>

Change 146655 on 2004/01/31 by mmantor@mmantor_xenos_linux_orl
<added include directory for crayola_enum.h and fixed vectpipe makefile to use env variables for dep dir>

Change 146654 on 2004/01/31 by mmantor@FL_mmantorLT_r400_win
<saved environment updates, problem with last checking>

Change 146651 on 2004/01/31 by mmantor@FL_mmantorLT_r400_win
<preparing for intrinsity release, updated makes to be compatible with multiple directories using SROOT and SBRANCH environment variables and cleaned up sp_vector.v for readability and removed unused signals, functionally equivalent>

Change 146535 on 2004/01/30 by llefebvr@llefebvr_r400_linux_marlboro
Adding Tom's changes to the emulator and SP testbench.

Change 146499 on 2004/01/30 by llefebvr@llefebvr_r400_linux_marlboro
Removing HW accureate flag

Change 146490 on 2004/01/30 by llefebvr@llefebvr_r400_emu_montreal
Removing numbers.h

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ATI 2048

LG v. ATI
IPR2015-00325

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Change 146478 on 2004/01/30 by lilefebvr@lilefebvr_r400_linux_marlboro

Fixing bad casting in the TB

Change 146091 on 2004/01/29 by lilefebvr@lilefebvr_r400_emu_montreal

Fixing SX buffer size management problem when EXPORT_BUFFER size > 128.

Change 146069 on 2004/01/29 by mmantor@mmantor_xenos_linux_orl

<updated makefile for standalone rand vector generator for vsp>

Change 146063 on 2004/01/29 by mmantor@FL_mmantorLT_r400_win

<added new files for last checkin>

Change 146062 on 2004/01/29 by mmantor@FL_mmantorLT_r400_win

<repartition more vector scaler pipe (vsp) into separte class to support lower level random test bench and intrinsity efforts.
moved all math from sq_alu.cpp to vsp.cpp>

Change 145619 on 2004/01/27 by lilefebvr@lilefebvr_r400_emu_montreal

Added VS_FETCH_DONE event and interface pulse to the CP for the PL_LIME_BUFFER early de-allocation of the vertex buffer.

Change 145089 on 2004/01/23 by lilefebvr@lilefebvr_r400_emu_montreal

Added event propagation to the SX (from the SQ). Need to tie it off in Xenos to the BC.

Change 143941 on 2004/01/20 by lilefebvr@lilefebvr_r400_linux_marlboro

- 1) Fixed Kill and waterfall issue (r400sq_killgt_01.cpp)
- 2) Fixed waterfall and mova co-issue problem (only in non-optimized mode).
- 3) Fixed SX dumpfile NAN sign propagation problem.

Change 143832 on 2004/01/19 by mmantor@FL_mmantorLT_r400_win

<updated the vectpipetest.exe do process interleaved threads, all opcodes and changed tb_vector back to simple design. All ops pass 400000 instruction random testing >

Change 143255 on 2004/01/15 by lilefebvr@lilefebvr_r400_emu_montreal

Added VS_FETCH_DONE control flow instruction.
Added VS_FETCH_DONE pulse to CP interface.
Placed the hooks for when the event will be created.

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Change 143096 on 2004/01/14 by hartogs@hartogs_xenos_linux_orl

Change to fix ferret dump file for VGT_SQ_Verts interface. This change passed regress_e and parts_lib_rel.pl.

Change 141868 on 2004/01/09 by lilefebvr@lilefebvre_laptop_r400_emu

Adding warnings for bad register settings on VS_CONST_BASE and VS_CONST_SIZE (sum must be < 512).

Change 141694 on 2004/01/08 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing DB_ALU_SIZE to say we need 2X the number of constants than highest use in a state.

Fixing VS_NUM_REG to say it is number of registers -1.
Fixing bug in predicated loop end where the jump back to the start of the loop wasn't taken.

Change 140866 on 2004/01/06 by lilefebvr@lilefebvre_laptop_r400_emu

This is the new alu code as provided by Tom. uses the new 27 bit adder.
Passed release_parts_lib, the emulator regression and the tb_sqpsx regression. also passed 4000000 vectors on the SP testbench.

Change 140078 on 2003/12/29 by lilefebvr@lilefebvr_r400_emu_montreal

Fixing flush to zero problem on MUL_PREV2.
Fixing _CONST abs modifier problem.

Change 139972 on 2003/12/29 by lilefebvr@lilefebvr_r400_emu_montreal

Reverting Tom's changes (try 3).

Change 139970 on 2003/12/29 by lilefebvr@lilefebvr_r400_emu_montreal

Reverting Tom's changes (try 2).

Change 139968 on 2003/12/29 by lilefebvr@lilefebvr_r400_emu_montreal

Reverting Tom's changes. They brake the random tests with LSB mismatches.

Change 139340 on 2003/12/23 by lilefebvr@lilefebvr_r400_emu_montreal

Fixed SX to send Nans on invalid positions.
Fixed waterfalloing on _CONST opcodes.
Added more debug info to SX->PA dumps.

Change 139084 on 2003/12/22 by lilefebvr@lilefebvr_r400_emu_montreal

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Adding more dump info in the SX->RB color dump.
Adding bug fixes in 27 bit adder path (not used).

Change 138986 on 2003/12/20 by mmantor@mmantor_xenos_linux_orl

<changed csim to only make one pass for param gen and gen index and write the dump files correctly, fixed a timing loop in pix tthread buffer >

Change 138150 on 2003/12/17 by lilefebvr@lilefebvr_r400_emu_montreal

Removed power of 2 specific wrapping to allow any size VTX and PIX RS.

Change 138102 on 2003/12/17 by lilefebvr@lilefebvr_r400_emu_montreal

Added programable VTX and PIX RS sizes (from state register).
Added programable SX export buffer sizes.
Added programable number of EA buffers in SX.
Added programable State export slots in SX.

Change 138048 on 2003/12/17 by lilefebvr@lilefebvr_r400_emu_montreal

The emulator now outputs an error message and exits upon bad usage of the address register.

Change 137942 on 2003/12/16 by lilefebvr@lilefebvr_r400_linux_marlboro

The testbench now compiles and runs on linux.

Change 137864 on 2003/12/16 by rramsey@rramsey_xenos_linux_orl

Change emulator so param cache reads for params not exported by the VS still show up in sq_sx_pcaddr.
Fix cf_resource_change logic in the cfs so it catches the clause boundary where a cf instr with only tex instr gets sent to the alu cfs.

Change 137313 on 2003/12/12 by mmantor@FL_mmantorLT_r400_win

<update for standalone vp test>

Change 137273 on 2003/12/12 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing another data scrmable on TRUNC scalar.
Modified MAX4v to match exactly numerics.
Modified MAX and Min opcodes to remove the subtract that could cause an LSB mismatch on the compare.
Fixed Nan on FLOOR scalar.

Change 137256 on 2003/12/12 by lilefebvr@lilefebvre_laptop_r400_emu

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This is fixing the Nan and data scramble on scalar TRUNC.

Change 137124 on 2003/12/12 by lilefebvr@lilefebvre_laptop_r400_emu

This is fixing the _CONST opcode failure. This is an SQ only change so it will not affect the SP testbench.

Change 137080 on 2003/12/11 by mmantor@FL_mmantorLT_r400_win

<tmp remove scalar prev opcodes for standalone testing>

Change 137063 on 2003/12/11 by lilefebvr@lilefebvre_laptop_r400_emu

Changing swizzles on CUBE instruction to exactly match SP.

Change 136833 on 2003/12/10 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing PRED_SET* opcodes to match HW behavior.
Fixing bad channel select on KILL* and MAX4 opcodes.

Change 136832 on 2003/12/10 by mmantor@FL_mmantorLT_r400_win

<added all ops and provide pred opcode swizzle changes and kill in the standalone vectpipetest>

Change 136727 on 2003/12/10 by mmantor@FL_mmantorLT_r400_win

<added all vector ops>

Change 136641 on 2003/12/10 by lilefebvr@lilefebvre_laptop_r400_emu

Fix for vector TRUNC +/- 0 clamping issues.

Change 136555 on 2003/12/09 by mmantor@FL_mmantorLT_r400_win

add VectPipeTest that test the vector pipe of the sp.

Change 136540 on 2003/12/09 by lilefebvr@lilefebvre_laptop_r400_emu

This should fix TRUNC nan failures and DOT2Add INF and NAN failures.

Change 136519 on 2003/12/09 by lilefebvr@lilefebvre_laptop_r400_emu

Another pass on the SP files to be more HW lookalike.

Change 136387 on 2003/12/09 by lilefebvr@lilefebvre_laptop_r400_emu

This is the new SP files more HW lookalike. Also fixes DOT2ADD SRCC swizzle issues.

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Change 136287 on 2003/12/08 by llefebvr@llefebvre_laptop_r400_emu

The TRUNC was rounding instead of clamping when dealing with negative numbers (LSB of mantissa only)

Change 135538 on 2003/12/04 by vromaker@vromaker_emul_r400_linux_marlboro

- updated vtx and pix thread buffer sizes

Change 135361 on 2003/12/04 by llefebvr@llefebvr_r400_emu_montreal

This should fix +/- problems in CUBE (501-502). And TRUNC to +0 issue (trunc_02).

Change 134820 on 2003/12/02 by llefebvr@llefebvr_r400_emu_montreal

Fixing the CUBE opcode. It was reading from SRCB.X while it should have read from SRCA.W.

Change 134536 on 2003/12/01 by llefebvr@llefebvr_r400_emu_montreal

Fixing bad swizzle in the case of a co-issued 2 operand scalar op along with a 3 operand vector op. Emulator was doing x and w while it should have been z and w.

Change 133958 on 2003/11/25 by llefebvr@llefebvr_r400_emu_montreal

There was a parenthesis problem in the if causing the fix for the +/- zero not to work on the Muladd. This is fixing it.

Change 133810 on 2003/11/25 by llefebvr@llefebvr_r400_emu_montreal

Fixing Add/MulAdd +/-0 problem in the emulator

Change 133619 on 2003/11/24 by llefebvr@llefebvr_r400_emu_montreal

Fixing additionnal control flow entry in the SQ dumps.

Change 133415 on 2003/11/21 by llefebvr@llefebvr_r400_emu_montreal

Fixed the wrong function for the +/- Zero bug on the Muladd engine.

Change 133358 on 2003/11/21 by llefebvr@llefebvr_r400_emu_montreal

Adding RETAIN_PREV opcode to the scalar engine as opcode 0x32 (last one).
Opcode has syntax:

RETAIN_PREV Rx; Where Rx is any register. To just retain the predicate vector use:
RETAIN_PREV Rx@;

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Fixed bad SRC selection for Tc in cube opcode.
Fixed +/- 0 problem in the multiply engine.

Change 133025 on 2003/11/20 by mearl@mearl_xenos_linux_orl

Fixed 2 primitive per clock interpolation bug in the emulator.

Change 132250 on 2003/11/17 by llefebvr@llefebvr_r400_linux_marlboro

Changing emulator and tests to meet with the new cube swizzles wich now are for SRCA zzxy (instead of zzys). Also change the assembler to accept the new swizzle code.

Change 131172 on 2003/11/10 by llefebvr@llefebvre_laptop_r400_emu

This fixes KILL* and PRED_SET_INV opcodes when working with denoms.

Change 130817 on 2003/11/07 by llefebvr@llefebvr_r400_emu_montreal

Fixing SUB NaN sign propagation.
Fixing PRED* flush of denoms to 0.
Also fixing +/-0 flush to zero problems in multiple other opcodes.

Change 130473 on 2003/11/06 by llefebvr@llefebvr_r400_emu_montreal

Now using SRCB.z for input argument of the Z channel ma computation when Z wins in a cube opcode.

Change 130441 on 2003/11/06 by llefebvr@llefebvr_r400_emu_montreal

Removal of duplicate code.

Change 130357 on 2003/11/05 by llefebvr@llefebvr_r400_emu_montreal

Fixing the CUBE opcode. Was broken because the emulator wasn't taking into account that SRCA and B had a different set fo swizzles.

Change 130281 on 2003/11/05 by llefebvr@llefebvr_r400_emu_montreal

Fixing SRCC valid GPR valid channel.
Putting the SERIAL on the right line the the cubic pixel shader program.

Change 130216 on 2003/11/04 by mmantor@FL_mmantorLT_r400_win

<changes to enable standalone vector pipe random testbench>

Change 130154 on 2003/11/04 by llefebvr@llefebvr_r400_emu_montreal

Updated DOT2ADD opcode to propagate NAN from SRC C as well.

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Change 130063 on 2003/11/04 by llefebvr@llefebvr_r400_emu_montreal

This should fix NAN sign propagation on the SUBs* opcodes (including SUB_PREV).

Change 129772 on 2003/11/03 by llefebvr@llefebvr_r400_emu_montreal

This fixes a problem were the dummy line in the SP->SX dump to signify a dump free done on memory export was places between phases 2 and 3 of an export instead of being between two exports. This change will only affect the TB_SQSPSX and TB_SX

Change 129642 on 2003/10/31 by llefebvr@llefebvr_r400_emu_montreal

New ExecuteVPInst function more flexible and hence more suitable to the testbench.

Change 129587 on 2003/10/31 by llefebvr@llefebvr_r400_emu_montreal

Instanciation of multiple memories in the emulator to allow for use of
SQ_DEBUG_MISC.DB_EN_MEMORY_X and DB_READ_MEMORY register fields.

Change 129454 on 2003/10/30 by llefebvr@llefebvr_r400_emu_montreal

This should make the CUBE opcode match exactly HW even if SRCA and B are different.

Change 129388 on 2003/10/30 by llefebvr@llefebvr_r400_emu_montreal

The emulator does not clamp the W channel anymore on a clamped CUBE operation. This is in order to mimmic HW behavior.

Change 129348 on 2003/10/30 by mearl@mearl_xenos_linux_orl

Added two primitive interpolation back in.

Change 129150 on 2003/10/29 by llefebvr@llefebvr_r400_linux_marlboro

Increasing VC mini count to 11_fifo_size +2.

Change 129117 on 2003/10/28 by danh@danh_xenos_linux_orl

Changed sq_sx_pcaddr.dmp generation.

Change 128927 on 2003/10/28 by llefebvr@llefebvr_r400_emu_montreal

Fixing precision problem in all PRED* vector opcode. Emulator was rounding.

Change 128679 on 2003/10/27 by llefebvr@llefebvr_r400_emu_montreal

This fixes an emulator dump problem related to VSISR address when vtx_count is on but VSR_continued if off.

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Change 128526 on 2003/10/24 by mearl@mearl_xenos_linux_orl

Took out two prim per clock to get regression to pass.

Change 128365 on 2003/10/24 by mearl@mearl_xenos_linux_orl

Added 2 primitive interpolation in SQ and SPI. Fixed a bug in sx_parameter_cache.
Fixed synthesis
bugs in SC.

Change 128179 on 2003/10/23 by llefebvr@llefebvr_r400_emu_montreal

This should fix the bad data type on a position free done in the sp->sx emulator dump file. This will only affect the TB_SQSP,TB_SQSPSX and TB_SX test benches as they are the only ones using this field.

Change 127742 on 2003/10/22 by llefebvr@llefebvr_r400_linux_marlboro

Removed the warnings from the sp->sx trackers and sx->sp.
Now emulator is always executing the scalar instruction even in the case of a 3 operand vector opcode. This is to match with random shaders.

Change 127044 on 2003/10/16 by llefebvr@llefebvre_laptop_r400_emu

There was a bug in the post steer valid bits sent to the TP and VC. This is fixing it.

Change 127009 on 2003/10/16 by llefebvr@llefebvre_laptop_r400_emu

This fixed the pixel auto counter bug in the interpolators.

Change 126957 on 2003/10/16 by llefebvr@llefebvre_laptop_r400_emu

Now pushing the PERFORMANCE_STOP events to the RS.
Fixed the predicate jump bug (wasn't reading the correct bit in the control flow instruction).

Change 126833 on 2003/10/15 by dclifton@dclifton_xenos_linux_orl

Small change to log result for numbers less than 1.0 to match HW.

Change 126691 on 2003/10/15 by dclifton@dclifton_xenos_linux_orl

Another timing related change. Changed twos comp to ones comp on log post-process (effects log of number less than 1.0). Aligned inputs to high precision pipeline to reduce muxing. Improved carriesave add of multiplier results. Regenerated math tables to reclaim precision and fix roll-over mismatches.

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Change 126641 on 2003/10/14 by lilefebvr@lilefebvre_laptop_r400_emu

Adding conditional_predicate_jmp and conditionnal_predicate_call. These are performance optimizations only and are not in use at the moment.

Change 126483 on 2003/10/13 by mearl@mearl_xenos_linux_orl

Fix One Prim Per Clock bug in sq_ptr_buff. Revert changes in sq_pix_ctl to make 2 prim interp changes easier. Put known primdata data on all quads across packer to iterator interface. Fix dumps for no_inc_pix_cnt signal.

Change 126362 on 2003/10/13 by rramsey@rramsey_xenos_linux_orl

Change sq_sp_interp dump so it contains all of the pass_count and wrap passes through the interpolator
Add spi_sp_tracker (enabled with ENABLE_SPI_TRACKER define)

Change 125881 on 2003/10/09 by lilefebvr@lilefebvr_r400_emu_montreal

Now doing a MAX on the vector and scalar MOVA opcodes to match HW.

Change 125617 on 2003/10/08 by lilefebvr@lilefebvr_r400_emu_montreal

Now passing the ROM data at the end of each line in the dump file. This is for RSP usage in testbenches.

Change 125370 on 2003/10/07 by mearl@mearl_xenos_linux_orl

Fixed the SQ bug when bad pipe exists before a good pipe. Also, updated the RT trackers in the SC testbench.

Change 125266 on 2003/10/07 by lilefebvr@lilefebvr_r400_emu_montreal

Added RSP fields to the dumps to allow for easier testbench integration.

Change 125023 on 2003/10/06 by lilefebvr@lilefebvr_r400_emu_montreal

Cleaned up the COND_PRED_EXECUTE and PRED_LOOP code in the emulator. No bug fixes here, just made the code easier to read and removed dead paths.

Change 124883 on 2003/10/03 by lilefebvr@lilefebvr_r400_emu_montreal

Fixing issue with NaNs and TRUNC,FLOOR,FRACT. Also, now SIN returns +/- 0 on input +/- 0.

Change 124851 on 2003/10/03 by lilefebvr@lilefebvr_r400_emu_montreal

Interpolation precision change to meet HW and timing.

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Change 124843 on 2003/10/03 by lilefebvr@lilefebvr_r400_linux_marlboro

These are register changes related to new arbitration policies for the ALU engine.

Change 124643 on 2003/10/02 by lilefebvr@lilefebvr_r400_emu_montreal

Fixing hang on pred_optimization test.

Change 124518 on 2003/10/02 by lilefebvr@lilefebvr_r400_emu_montreal

This implements the cond_pred optimization in the emulator in order to match HW in the number of clause boundaries.

Change 124434 on 2003/10/01 by mmang@mmang_xenos_linux_orl

1. Turned on 3 simds in emulator (sc_interp.cpp, sq_block_model.cpp, and user_block_model.cpp).
2. Turned on 3 simds in rtl (sc_packer.v, tb_sqsp.v, and vgt.v).
3. Fixed bug in chip_vc tree to get SQ_VC_simd_id and TC_VC_simd hooked up correctly.
4. Fixed bug in sc_packer.v related to having a 2 bit simd_id_sel.

Change 124011 on 2003/09/30 by lilefebvr@lilefebvr_r400_emu_montreal

Autoreg change to remove the _no_stalls opcodes and replace them with _pred_clean. Documentation only should not be a functional change. I had to remove some references to _no_stall in the emulator code as well.

Change 123879 on 2003/09/29 by lilefebvr@lilefebvr_r400_emu_montreal

Fixing some RSP problems on the Sq->TP/VC interfaces. These problems only show up when RSP is turned on (it is currently off).

Change 123610 on 2003/09/26 by lilefebvr@lilefebvr_r400_emu_montreal

Trying to FIX bad NAN handling on the scalar pipe.

Change 123497 on 2003/09/26 by lilefebvr@lilefebvr_r400_emu_montreal

Added the SP->SX RSP interface. Also added post_steered bits to all interfaces for tracker purposes only (not HW). These are the valid bits of the SPs AFTER the data was steered to go to the RSP.

Change 123278 on 2003/09/25 by lilefebvr@lilefebvr_r400_emu_montreal

Added TP redundant pipe. Also renamed all redundant pipes to use SP instead of SQ.

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Change 123130 on 2003/09/24 by lilefebvr@lilefebvr_r400_emu_montreal

First pass at redundant pipe in the emulator. For now, only on the VC->SQ and SQ->VC interfaces.

Change 123082 on 2003/09/24 by mearl@mearl_crayola_linux_orl

tb files updated for ONE_PRIM_PER_CLOCK, bug fix in interpolators for ONE_PRIM_PER_CLOCK

Change 122815 on 2003/09/23 by lilefebvr@lilefebvr_r400_linux_marlboro

Added more perf counters to the SQ to account for SIMD2-3.

Change 122728 on 2003/09/23 by lilefebvr@lilefebvr_r400_emu_montreal

Made the autocount write to all channels like the HW does. Auto count in X, 0 in all other channels.

Change 122683 on 2003/09/23 by mearl@mearl_crayola_linux_orl

One primitive per clock changes in the back of the SC and front of the SQ. Right now, the ONE_PRIM_PER_CLOCK define in header.v and SC_SQ_interface.v are needed for this change. Will update this to ONEPPC, since this already exists in header.v. Also, the sim.cfg file does not have an ifdef, so is hardcoded to one prim per clock.

Change 122677 on 2003/09/23 by dclifton@dclifton_xenos_linux_orl

Changed initial twos comp on exp opcode to ones comp for timing improvement in hardware.

Change 121928 on 2003/09/17 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing multiple SIMD related stuff.

Change 121897 on 2003/09/17 by dclifton@dclifton_crayola_linux_orl

Change in HW structure to improve timing

Change 121632 on 2003/09/16 by lilefebvr@lilefebvre_laptop_r400_emu

Made the XYs be sent as floating point numbers instead of fix point so you can use them directly in the shader. Also modified regress_e tests that this change broke. Also added register fields for SIMD memory control.

Change 120731 on 2003/09/11 by lilefebvr@lilefebvr_r400_emu_montreal

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I have removed the predication optimization on waterfall from the emulator. The HW does not do it. Lines are marked with @@ OPTIMIZE so we can know what was removed in order to put it back if needs be.

Change 120546 on 2003/09/10 by lilefebvr@lilefebvr_r400_emu_montreal

Adding the scalar engine change. This change is currently disabled. To enable, change line # 105 of sclarfunc.h from:

```
__int64 ComputeMantissa6X(unsigned int point, unsigned int slope,
    unsigned int slope_derivative,
    unsigned int sloped_derivative,
    unsigned int delta, unsigned int ddelta,
    unsigned int dddelta, McParam2 *params,
    int gen_type, int new_struct = 0);
```

to:

```
__int64 ComputeMantissa6X(unsigned int point, unsigned int slope,
    unsigned int slope_derivative,
    unsigned int sloped_derivative,
    unsigned int delta, unsigned int ddelta,
    unsigned int dddelta, McParam2 *params,
    int gen_type, int new_struct = 1);
```

Change 120264 on 2003/09/09 by lilefebvr@lilefebvr_r400_emu_montreal

Now writing the channel valid mask in hex in the dumps.

Change 120058 on 2003/09/08 by chammer@chammer_r400_win

Changed width of dealloc from 3 to 4 bits since it can now accumulate to 8 in 1 prim per clock mode.
Load prim data into 'bad pipes' to match hardware.

Change 119590 on 2003/09/05 by lilefebvr@lilefebvre_laptop_r400_emu

Rearranged the functions in the shader pipe for SP testbench purposes. No functional changes per say.

Change 118543 on 2003/08/28 by lilefebvr@lilefebvr_r400_emu_montreal

Adding thread type to SX interfaces for memory export validation.
Also made sure to invalidate the Parameter cache interface reads whenever generating a parameter.

Change 118385 on 2003/08/27 by lilefebvr@lilefebvr_r400_emu_montreal

Added the thread Id identifier to interfaces for tracking purposes of memory export tests.

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Change 118345 on 2003/08/27 by llefebvr@llefebvr_r400_emu_montreal

Missing a file in the changelist.

Change 118332 on 2003/08/27 by llefebvr@llefebvr_r400_emu_montreal

Massive changelist of mostly non-functionnal stuff with two exeptions:

- 1) Now checking for INF in the DOT path and doing as per the numerics spec.
- 2) Changed the VC mini instruction count limit from 16 to 32 as instructed by Brian Buchner.

The rest of the changelist is the first pass to correct the invalid channels in the GPRs in order for them not to be compared. This is mostly only adding bits at the end of the dumps for the trackers to have the information needed to know what's valid and what's not. Interfaces changed are:

- 1) Internal SP GPR
- 2) SP->SX
- 3) SX->SP
- 4) SX->PA
- 5) SX->RB
- 6) SX->Interp (PC data)
- 7) SP interpolators

Change 118211 on 2003/08/26 by rramsey@rramsey_crayola_linux_orl

changes to get fake_last set correctly in tp_sqsp.dmp

Change 117786 on 2003/08/22 by llefebvr@llefebvre_laptop_r400_emu

The loop index clamping wasn't done correctly as I was trying to force negative values on unsigned variables.

Change 117590 on 2003/08/21 by llefebvr@llefebvre_laptop_r400_emu

I broke pretty much all waterfall tests with my previous checkin. This is a fix.

Change 117545 on 2003/08/21 by llefebvr@llefebvre_laptop_r400_emu

This is an attempt to fix the bug 2515. I made sure a pixel wasn't considered in subsequent passes if it was written. I will need sunshine to test it as I can't run the app.

Change 117283 on 2003/08/20 by llefebvr@llefebvre_laptop_r400_emu

For parameter cache (interpolators) exports, I was using a different function to output the results to a file. I did not bother opening up the file in this function as I suspected the position export function would open it for me. Obviously, if you export position last this doesn't work. I

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modified the function to check for file open and open it if not. This fixes the numerous bugs the compiler guys were seeing when exporting position last. This bug would only show up if dump files are activated.

Change 117162 on 2003/08/19 by llefebvr@llefebvre_laptop_r400_emu

Fixing CUBE instruction by move the *2.0 from red to blue.

Change 117149 on 2003/08/19 by llefebvr@llefebvre_laptop_r400_emu

Now clamping to -256 if overflow of the loop index register.

Change 116305 on 2003/08/13 by chammer@chammer_r400_win

Added one prim per clock to SC. This code is ifdef'd and can be enabled by uncommenting out `//#define ONE_PRIM_PER_CLK` in `gfx/sc/sc_types.h` as well as in `gfx/sq/sc_sq.h`

Change 115549 on 2003/08/08 by llefebvr@llefebvr_r400_emu_montreal

Now writing DEADDEAD in the parameter caches at alloc time.
Any DEADDEAD pixel or vertex will go thru CLAMP unchange is it is a DEADDEAD.

Change 115328 on 2003/08/07 by llefebvr@llefebvr_r400_emu_montreal

The HW has a 26 bits normalizer so I increased the emulator's precision to match. This fixes `r400sq_ripple_01.cpp`. I had to re-goldenize 1 test in the `regress_e` suite.

Change 115032 on 2003/08/05 by grayc@grayc_crayola2_linux_orl

added back Laurent changes for sx performance counters
modified `sx.v` for new performance register names

Change 114997 on 2003/08/05 by llefebvr@llefebvr_r400_emu_montreal

Forgot to flush denorms prior to check for `== 0`. This was causing some tests to mismatch on the `SX->SP` parameter sub interface.

Change 114116 on 2003/07/31 by grayc@grayc_crayola2_linux_orl

backing out changes for SX Perf Counters

Change 114066 on 2003/07/30 by llefebvr@llefebvr_r400_emu_montreal

Aligned correctly the `sq_sp_interp.dmp`
Made a small change to generate -0 in the `param_bub` engine of the SX when both operands are the same and SRCa is negative. This is to match the HW algorithm.

Change 114014 on 2003/07/30 by llefebvr@llefebvr_r400_emu_montreal

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Made the SX performance counters unified in one SX. Also added some performance counters to monitor the `SX->RB` performance. All of these are non-windowed yet.

Change 113865 on 2003/07/29 by llefebvr@llefebvr_r400_emu_montreal

This toggles the emulator into a single/multiple SIMD engine. Can toggle on/off by `undef/def` line 41 of `sq_block_model.cpp` (`#define ENABLE_MULTIPLE_SIMDS`). Currently OFF.

Change 113818 on 2003/07/29 by llefebvr@llefebvr_r400_emu_montreal

Now routing SIMD id to the SQ and TP dumps.
Added 4 GPR banks to support up to 4 SIMD engines all having the same GPR address but different data.

Added 4 allocation engines to generate these addresses.
The number of SIMD engines is user defined it is a `#define` in `sq/user_block_model.h`, current setting is 4 but only 2 are used as the SC and VGT never send more than that for now.

Change 113562 on 2003/07/28 by llefebvr@llefebvr_r400_emu_montreal

Cleaned and Fixed memory exports.
Made the resource register pick the VC if a vertex thread is entered.

Change 112593 on 2003/07/23 by llefebvr@llefebvre_laptop_r400_emu

Fixing infinite loop problem on 0 count `exec_ends` because of not setting correctly the end of program flag in the SQ.

Change 112528 on 2003/07/22 by hartogs@fl_hartogs2

Made the `simd_id` field between the VGT and the SQ two bits for up change to three or more SIMD sets.

Change 112332 on 2003/07/22 by llefebvr@llefebvre_laptop_r400_emu

I had the old output order for cube: `ma,faceid,sc,tc`. I changed it for the new one: `tc,sc,ma,faceid`.

Change 112254 on 2003/07/21 by llefebvr@llefebvre_laptop_r400_emu

This is the CUBE opcode change that takes into account the recent HW change. I also modified one test case that was wrongfully picking W as the FACEID (it is Y).

Change 111835 on 2003/07/18 by llefebvr@llefebvr_r400_emu_montreal

Splitting the ALU CONSTANT perf counters into SIMD0 and 1. I have added the SIMD1 version of the counters at the end of the enumeration not to disturb again the current order.

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Change 111444 on 2003/07/16 by llefebvr@llefebvr_r400_emu_montreal

Added sin/cos corner cases.

Change 111399 on 2003/07/16 by llefebvr@llefebvr_r400_emu_montreal

Fixing HOS anomaly because of bad SIMD setting in the SQ.
Adding the auto-generated parameter in the assert for too many parameters in the RT SQ.

Change 111005 on 2003/07/14 by llefebvr@llefebvr_r400_emu_montreal

Changing the RT assert to `<=` instead of `<`

Change 111001 on 2003/07/14 by llefebvr@llefebvr_r400_emu_montreal

Adding the SIN/COS opcodes in the emulator.

Change 110861 on 2003/07/14 by llefebvr@llefebvr_r400_emu_montreal

SQ now only reading TP interface data when `Phase == 3 AND TP_SQ_VALID`. Also added an assert when trying to interpolate more than 4 parameters in RT mode.

Change 110814 on 2003/07/14 by jhoule@jhoule_r400_win_lt

Added separate valids for TP_SQ and TP_SP data.
This will eventually allow simpler HiColor data return from TP.

Change 110319 on 2003/07/10 by llefebvr@llefebvr_r400_emu_montreal

Wasn't setting the VC counters correctly in the SQ. Was causing a hang whenever an app would use a mini-fetch.

Change 109729 on 2003/07/08 by llefebvr@llefebvr_r400_emu_montreal

Fixed the pixel counter in the SQ. This value is not used yet.

Change 108547 on 2003/06/30 by lseiler@lseiler_r400_win_marlboro

Push out MemExports before processing certain events.

Change 108285 on 2003/06/27 by llefebvr@llefebvre_laptop_r400_emu

Swapping write priority to a GPR. Was vector has priority over scalar, now IS scalar has priority over vector.

Change 108154 on 2003/06/26 by hartogs@fl_hartogs2

Hopefully fixed VGT alloc/dealloc for multi-SIMD vector sets

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Added simd_id fields to vgt_sq interface and vgt_ccgen interface
Put pipedisable comments into several dump files.
Put an "Assert(0)" into the sq_block_model.cpp to prevent access violation.

Change 107916 on 2003/06/25 by lilefebvr@lilefebvre_laptop_r400_emu

Made all the necessary changes for the color buffers to be enlarged without actually enlarging them so I don't break anything.

Change 107624 on 2003/06/24 by lilefebvr@lilefebvre_laptop_r400_emu

Now enlarging valid bit mask to be quad aligned for pixel vectors (in order to get LOD right to the TP).

Change 107575 on 2003/06/24 by lilefebvr@lilefebvre_laptop_r400_emu

Added back pressure ability from the VC to the SQ using the Mega and mini counters.

Change 107515 on 2003/06/23 by mzini@mzini_r400_win

Added vc_sq_sync interface

Change 107512 on 2003/06/23 by mzini@mzini_r400_win

Created VC_SQ_Sync interface

Change 107026 on 2003/06/19 by lilefebvr@lilefebvr_r400_linux_marlboro

1) Added a guard bit to the parameter sub engine of the SX in both the emulator and HW this was causing a failure on a WQL test.
2) Fixed zero detection problem in parameter_sub engine of the HW were an explicit 1 was added all the time even when the number was 0.0. This was causing r400sx_wrapper_01.cpp to fail (this is a test that I wrote to duplicate the WQL test that was failing in order to run it on HW).

Change 106833 on 2003/06/18 by lilefebvr@lilefebvr_r400_emu_montreal

Added loop index clamping to range -256,255

Change 106809 on 2003/06/18 by lilefebvr@lilefebvr_r400_emu_montreal

Was using the textureCF machine for the VC incorrectly. Caused the VC to fail.

Change 106526 on 2003/06/17 by lilefebvr@lilefebvr_r400_emu_montreal

Was incorrectly reading uninitialized data.

Change 106470 on 2003/06/17 by rramsey@rramsey_crayola_linux_orl

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Change interpolators so sx_sp_pcdump is not written for param_gen or gen_index passes (param cache is not actually read for those)
Clean up sqsp_interp dump

Change 106404 on 2003/06/16 by lilefebvr@lilefebvr_r400_emu_montreal

Initializing GPRS with DEADDEAD for ferret tracking purposes. Each thread is initialized with this at GPR allocation time.

Change 106347 on 2003/06/16 by lilefebvr@lilefebvr_r400_emu_montreal

Made the wrapping test >= instead of >

Change 106339 on 2003/06/16 by lilefebvr@lilefebvr_r400_emu_montreal

Fixing SQ->VC bad header in dump.
Fixing bad mova clamping.

Change 106263 on 2003/06/16 by mzini@mzini_r400_win

Removed y and z from the index field in sq_vc.dump file AGAIN!

Change 106093 on 2003/06/13 by lilefebvr@lilefebvr_r400_emu_montreal

Emulator now clamps the address register to the range -256...255.

Change 106000 on 2003/06/13 by lilefebvr@lilefebvr_r400_emu_montreal

Fixing SQ->VC dump to be correctly aligned.

Change 105865 on 2003/06/12 by lilefebvr@lilefebvr_r400_emu_montreal

Fixing bad write mask setup on waterfall of lots of vertexes. Also enable waterfall_optimize on exports.

Change 105795 on 2003/06/12 by lilefebvr@lilefebvr_r400_emu_montreal

Fixing waterfaling on a MOVA instruction.

Change 105772 on 2003/06/12 by lilefebvr@lilefebvr_r400_emu_montreal

Fixed a bug that caused generation of INF in the interpolators when param 0 was INF regardless of param 1 and 2. Also realigned the sq_sp_interp.dump.

Change 105645 on 2003/06/11 by lilefebvr@lilefebvr_r400_emu_montreal

There was a problem in the way the masks were set when waterfaling. If multiple vertexes were using the same constant on the second pass, only one of them was set.

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Change 105336 on 2003/06/10 by lilefebvr@lilefebvr_r400_emu_montreal

Waterfaling wasn't tied to write mask of GPRs hence creating corruption when running in optimized mode.

Change 105295 on 2003/06/10 by lilefebvr@lilefebvr_r400_emu_montreal

Bad initialization for optimized waterfaling that caused linux to loop indefinitely.

Change 105276 on 2003/06/10 by lilefebvr@lilefebvr_r400_emu_montreal

Changing the order of waterfaling to match HW on the interface. Emulator was doing LSB->MSB while HW was doing MSB->LSB.

Change 105001 on 2003/06/09 by lilefebvr@lilefebvr_r400_emu_montreal

Added waterfall write masks to the emulator.
Fixed the ALU instruction dump file to work with waterfall.

Change 104944 on 2003/06/09 by lilefebvr@lilefebvr_r400_emu_montreal

Modified the emulator to generate the same number of passes than the HW on a waterfall instruction. Only the number of passes is correct at this point not the write mask.

Also corrected 2 MOVA tests where the address register wasn't loaded correctly before use.

Change 104090 on 2003/06/04 by lilefebvr@lilefebvr_r400_emu_montreal

Made 3 different counters for fetch, cst and instructions.

Change 104059 on 2003/06/04 by lilefebvr@lilefebvr_r400_emu_montreal

Added a RT only instruction shader dump.

Change 103964 on 2003/06/04 by lilefebvr@lilefebvr_r400_emu_montreal

- 1) Adding normalization stage to the SX
- 2) Fixing unitialized loop variables in the SQ.
- 3) Fixing RT/Normal constant and instruction loads (arbitrated on a 32 bit basis now)
- 4) Fixing cylindrical wrap problem in SX
- 5) Added performance counters for SIMD0/1

Change 103846 on 2003/06/03 by rramsey@rramsey_crayola_linux_orl

Fix a problem with the control flow dump logic that was causing zero-count execs to be left out of the file

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Change 103806 on 2003/06/03 by rramsey@RRAMSEY_P4_r400_win

fix comment line in SqSpInterp_Dump

Change 103675 on 2003/06/02 by mzini@mzini_r400_win

Added simd_id to sq_vc and vc_sq dumps

Change 103552 on 2003/06/02 by lilefebvr@lilefebvr_r400_emu_montreal

Added SIMD ID to TP interfaces.

Change 103230 on 2003/05/30 by mzini@mzini_r400_win

Added simd_id to SQ_VC and VC_SQ

Change 101734 on 2003/05/20 by lilefebvr@lilefebvre_laptop_r400_emu

fixing scalar trunc problem.

Change 100741 on 2003/05/13 by lilefebvr@lilefebvr_r400_emu_montreal

Fixing range underflow check of GPRs when using loop indexing and negative step.

Change 100446 on 2003/05/12 by lilefebvr@lilefebvr_r400_emu_montreal

was not checking for < BASE for constant indexing range checking.

Change 100178 on 2003/05/09 by lilefebvr@lilefebvr_r400_emu_montreal

bug with the VC arbiter.

Change 99938 on 2003/05/08 by hwiese@fl_hwiese_r400_win

SC, SQ, & VGT Emulator Update:
- Added shader pipe disable support for multiple SIMDs

Change 99865 on 2003/05/08 by lilefebvr@lilefebvr_r400_emu_montreal

Fog changes: Serialized fog in the SP after the default settings for exports.
Fixed assembler to put 0 in scalar write mask for BW compatibility with od shaders.

Change 99556 on 2003/05/07 by lilefebvr@lilefebvr_r400_emu_montreal

fixing the RT assertion.

Change 99443 on 2003/05/06 by lilefebvr@lilefebvr_r400_emu_montreal

Added an assert for bad constant registry setting regarding RT.

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Change 99140 on 2003/05/05 by lseiler@lseiler_r400_win_marlboro

Eliminate two vcc warning messages

Change 98355 on 2003/04/30 by hwiese@fl_hwiese_r400_win

ROM/GFX Emulator Update:

- 1) Removed register fields from ROM_BAD_PIPE_DISABLE_REGISTER
 - DISABLE_SP_VTX
 - DISABLE_SP_PIX
- 2) Added register ROM_SIMD_PIPE_DISABLE_REGISTER with fields
 - + DISABLE_SIMD0_VTX
 - + DISABLE_SIMD0_PIX
 - + DISABLE_SIMD1_VTX
 - + DISABLE_SIMD1_PIX
 - + DISABLE_SIMD2_VTX (reserved for future use)
 - + DISABLE_SIMD2_PIX (reserved for future use)
 - + DISABLE_SIMD3_VTX (reserved for future use)
 - + DISABLE_SIMD3_PIX (reserved for future use)
- 3) Fixed bug in ROM block where post write trigger function for ROM_BAD_PIPE_DISABLE_REGISTER was not named correctly in rom.mblk files causing function to never be called
- 4) Added post write trigger function for new register ROM_SIMD_PIPE_DISABLE_REGISTER
- 5) Updated PA, SC, SQ, and VGT blocks to use new SIMD0 fields when determining bad and/or disabled pipes
- 6) Added "TO DO" comment where SIMD1 logic needs to be added
- 7) Modified InitRomStraps() to use ROM.ROM_BAD_PIPE_FUSE_REG.devForce() to set the LASER_FUSES rather than ROM.ROM_BAD_PIPE_FUSE_REG.write() since this is a read-only register without a default setting

Change 98272 on 2003/04/30 by llefebvr@llefebvr_r400_emu_montreal

This should fix Ken's bug. Ken please confirm by running this in your sand box...

Change 97570 on 2003/04/25 by llefebvr@llefebvre_laptop_r400_emu

Added SIMD1_DISABLE register field.
Added MEMORY_READ register field.
Added perf control to control performance dumps.
Fixed the overwrite I made in cf_machine.cpp with previous checkin.

Change 97404 on 2003/04/24 by llefebvr@llefebvre_laptop_r400_emu

Making the SX more HW alike for memory exports.
Fixing the wrapping bug in the control flow machine (related to jumping in the instruction store).

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Change 97321 on 2003/04/24 by mmantor@FL_mmantorLT_r400_win

<fixed a bug with sx_sp_pcddata.dmp. The emulator was making the incorrect number of passes to do interpolation>

Change 97064 on 2003/04/23 by llefebvr@llefebvre_laptop_r400_emu

Added NO_SERIAL register.
Changed pred_set opcodes to use W instead of X (may break some SQ tests that will have to be fixed).

Change 97003 on 2003/04/22 by llefebvr@llefebvre_laptop_r400_emu

Added end of shader markers to shaders dump.

Change 96907 on 2003/04/22 by llefebvr@llefebvre_laptop_r400_emu

Added comments to SX dumps.
Added VC dumps.
Made emulator always stop on serial so it is deterministic.

Change 96369 on 2003/04/18 by llefebvr@llefebvr_r400_emu_montreal

adding some more code t the VC.
Fixed thread going always back on serial.

Change 96001 on 2003/04/16 by frising@frising_r400_win_marlboro

Move computation of a_mul_b_exp_flip_predict before a_mul_b_exp in muladd. This was the original intention and I suspect a copy/paste error at some point. In any case, it should not change the results. I checked with AndyG and apparently the HW is already doing this.

Change 95739 on 2003/04/15 by llefebvr@llefebvr_r400_emu_montreal

added the new performance counters in the SQ.

Change 95669 on 2003/04/15 by llefebvr@llefebvr_r400_emu_montreal

New dumps with shaders and number of vertexes + pixels per shader. Neede to validate load on the R500.

Change 95601 on 2003/04/15 by llefebvr@llefebvr_r400_emu_montreal

The SQ now sends data to the VC. A dump: sq_vc.dmp has also been added for test bench purposes.

Change 95456 on 2003/04/14 by llefebvr@llefebvr_r400_emu_montreal

Added the SQ->VC interface and VC->SQ interface.

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Change 95216 on 2003/04/11 by llefebvr@llefebvr_r400_emu_montreal

Added a VC empty shell. Need to add the interface to it. SQ is ready.

Change 94942 on 2003/04/10 by llefebvr@llefebvr_r400_emu_montreal

changed the comment name for free_done_position to FDP

Change 94940 on 2003/04/10 by llefebvr@llefebvr_r400_emu_montreal

Adding pos_free_done to the sp_sx.dmp for the SX testbench.

Change 94875 on 2003/04/10 by llefebvr@llefebvr_r400_emu_montreal

Sx now sends R400_NANs to the PA when VS_EXPORT_MODE == 7.
Added number of scalar operands to SP_in_dump.

Change 94729 on 2003/04/09 by llefebvr@llefebvr_r400_emu_montreal

Added the bit necessary to add the VC engine. Also modified the control flow instruction.

Change 94688 on 2003/04/09 by grayc@grayc_crayola2_linux_orl

correct filename

Change 94344 on 2003/04/07 by grayc@grayc_crayola2_linux_orl

move appending of \$TestPath into dump_point class for the rb,sx,sq,sc blocks

Change 94334 on 2003/04/07 by llefebvr@llefebvr_r400_emu_montreal

Fixing small discrepancies in the SP dumps. I also had a bad addressing on constants when using constant 0 on SRC B. Please Gang initiate a driver emulator drop and confirm this is fixed indeed.

Change 93911 on 2003/04/04 by llefebvr@llefebvr_r400_emu_montreal

I was not pushing the IP to and from the stack in the right order whenever moving to and from the RS. Also, now using a single table line in the SX export table when doing mem-exports and finxing additional entries in CF dump.

Change 93644 on 2003/04/03 by llefebvr@llefebvr_r400_emu_montreal

added channel masking to the SX.

Change 93506 on 2003/04/02 by llefebvr@llefebvr_r400_emu_montreal

adding jump/call address to the dump.

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Change 93480 on 2003/04/02 by llefebvr@llefebvr_r400_emu_montreal

There was an error in the dump comment line for number of bits in EXEC subword.

Change 93403 on 2003/04/02 by llefebvr@llefebvr_r400_emu_montreal

Adding CONTEXT_DONE events to the RS (not sending them to the CP).
Fixing the problem with the SP->SX interface (was making 1 too many transfert because of the dummy line inserted in the dump).

Change 92966 on 2003/03/31 by llefebvr@llefebvr_r400_emu_montreal

Added the dummy free done line in case there is no export as a last clause instruction.

Change 92927 on 2003/03/31 by llefebvr@llefebvr_r400_emu_montreal

removing the | from the CF,ALU and RS dumps, it confuses the PLI routines.

Change 92640 on 2003/03/28 by llefebvr@llefebvre_laptop_r400_emu

Added the instruction store read address to the CF instruction and ALU dumps.

Change 92630 on 2003/03/28 by llefebvr@llefebvre_laptop_r400_emu

Fixing MAX4 problem with NANs.

Change 92626 on 2003/03/28 by llefebvr@llefebvre_laptop_r400_emu

Fixing the write enables of the SP dumps.

Change 92563 on 2003/03/28 by llefebvr@llefebvre_laptop_r400_emu

Added number of operands in the SP dumps.

Change 92409 on 2003/03/27 by llefebvr@llefebvre_laptop_r400_emu

Fixing the sign extention.

Change 92390 on 2003/03/27 by llefebvr@llefebvre_laptop_r400_emu

fixing more alloc problems with events that were hanging SQ and SC tests.

Change 92227 on 2003/03/26 by llefebvr@llefebvre_laptop_r400_emu

This should fix that. Let me know.

Change 92184 on 2003/03/26 by llefebvr@llefebvre_laptop_r400_emu

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The SQ was not allocating in order for colors. This was causing the SX to misbehave. This is fixed.

Change 91585 on 2003/03/21 by llefebvr@llefebvr_r400_emu_montreal

Fixing wrong aluld number in arbiter.
Fixing mem exports some more.

Change 91363 on 2003/03/20 by llefebvr@llefebvr_r400_emu_montreal

flipping mask bit order

Change 91261 on 2003/03/20 by llefebvr@llefebvr_r400_emu_montreal

Major changes to the SQ/SX. Modified the SX to accept out of order exports and made it closer to HW. Also now matching the HW interface for allocs/deallocs in the SX.

Change 91256 on 2003/03/20 by llefebvr@llefebvre_laptop_r400_emu

fixing predication bug

Change 91169 on 2003/03/20 by rramsey@RRAMSEY_P4_r400_win

fix typos in two dump file names (pix/vtx was swapped for control flow dumps)

Change 90027 on 2003/03/13 by llefebvr@llefebvr_r400_emu_montreal

Fixed the Nan Check.
Added more fields to sp_sx.dmp and sx_sq_addr.dmp for Sx testbench.

Change 89938 on 2003/03/13 by rramsey@RRAMSEY_P4_r400_win

swap order of d0,1,2 in dump header

Change 89768 on 2003/03/12 by llefebvr@llefebvr_r400_emu_montreal

changing name export to Export because of name clash on linux.

Change 89716 on 2003/03/12 by llefebvr@llefebvr_r400_emu_montreal

SP data dumps.

Change 89381 on 2003/03/10 by llefebvr@llefebvr_r400_emu_montreal

Added dumps for ALU instructions.

Change 89037 on 2003/03/07 by llefebvr@llefebvr_r400_emu_montreal

fixing dumps and memory export test

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Change 89017 on 2003/03/07 by llefebvr@llefebvr_r400_emu_montreal

Modifying dumps not to exceed 32 bits/col (PLI need)

Change 89007 on 2003/03/07 by llefebvr@llefebvr_r400_emu_montreal

Added control flow instruction dumps.

Change 88898 on 2003/03/06 by rramsey@RRAMSEY_P4_r400_win

Add new block level dumps to SQ for the vertex input controller (enabled with SqDump >1)

sq_sp_vistr_wr.dmp - writes to staging registers
sq_sp_vec_gpr.dmp - reads from staging regs/writes to gprs
sq_vec_gpr_req.dmp - gpr allocate requests from the vertex controller

Change 88859 on 2003/03/06 by llefebvr@llefebvr_r400_emu_montreal

fixing 2D bug due to GPR addressing change to match HW.

Change 88737 on 2003/03/06 by grayc@grayc_crayola_linux_orl

changes for dump files

Change 88687 on 2003/03/05 by llefebvr@llefebvr_r400_emu_montreal

Fixing +/-0 bug in emu and type setting.

Change 88554 on 2003/03/05 by llefebvr@llefebvr_r400_emu_montreal

More SQ RS dumps.

Change 88079 on 2003/03/03 by llefebvr@llefebvre_laptop_r400_emu

Changed vertex GPR addressing to match HW exactly.

Change 87922 on 2003/03/01 by mmantor@FL_mmantorLT_r400_win

1. fixed a texture wrap bug.
2. fixed an sc hang for large vertices with lots of primitives culled where more than the max parameter cache space (2.5 times) could be required to operate. Put a fix that forces a partial fill of any vector being assembled if the previous vector had more than 4 outstanding deallocates out and a new vector (fpos) arrives at the packer input. This forces all the de-allocates to happen on the partial fill vector which will free pc space and allow the final vertex vector to process and etc. Added outstanding deallocate count to dump files sc_sq and sc_pix_vect_grp_out

Change 87326 on 2003/02/27 by llefebvr@llefebvre_laptop_r400_emu

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Added dumps for the reservation stations.

Change 87112 on 2003/02/26 by llefebvr@llefebvre_laptop_r400_emu

added predicate (export write mask) to the SP->SX interface. Also added them in emulator dump.

Change 87027 on 2003/02/26 by llefebvr@llefebvre_laptop_r400_emu

Fixed Trunc.

Change 86925 on 2003/02/25 by llefebvr@llefebvre_laptop_r400_emu

modified dealloc code slightly to improve debugging of deallocs.

Change 86890 on 2003/02/25 by llefebvr@llefebvre_laptop_r400_emu

Fixing Inf and DOTs.
Aligning RT_CONSTANTS with cleaner register boundaries.

Change 86634 on 2003/02/24 by mmantor@FL_mmantorLT_r400_win

added thread id to sp->sx interface dump
fixed sx to sp parameter data dump
added sq to sx pc_ptr dump
added texture cylindrical wrap control to sq

Change 86257 on 2003/02/22 by mmantor@FL_mmantorLT_r400_win

Set ClampedAddr = Addr before clamping so if no reason to clamp the address is defined. Stops emulator from crashing.

Change 86096 on 2003/02/21 by llefebvr@llefebvr_r400_emu_montreal

That should do it. I wasn't carefull enough about the clamping rules to the GPRs when the pointer was in fact used to read constants...

Change 85184 on 2003/02/19 by llefebvr@llefebvr_r400_emu_montreal

fixed clamping of -0 to +0.

Change 85039 on 2003/02/18 by llefebvr@llefebvr_r400_emu_montreal

Flushing denorms to 0 is now done on all three parameters (I forgot to do it on P0 in the emulator).

Change 84940 on 2003/02/18 by llefebvr@llefebvr_r400_emu_montreal

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Ex. 2048 --- R400 Sequencer Emulator FH --- folder_history

Fixing clamping problem in the emulator.

Change 84802 on 2003/02/17 by llefebvr@llefebvr_r400_emu_montreal

Code cleanup and fixing the problem with INF on the recip opcode.

Change 84554 on 2003/02/14 by llefebvr@llefebvr_r400_emu_montreal

fixing the clamp of the NANs to 0 and 1. They now go thru the clamp modifier unchanged.

Change 84487 on 2003/02/14 by llefebvr@llefebvr_r400_emu_montreal

repairing instruction count perf counters.

Change 84204 on 2003/02/13 by llefebvr@llefebvr_r400_emu_montreal

DOT now returns an R400_NAN instead of the NAN it was provided with.

Change 83625 on 2003/02/11 by llefebvr@llefebvr_r400_emu_montreal

there was a typo in the interface. The context id was sent as the channel mask and vice versa to the TP...
Also added PC exports to the sp_sx.dmp file for the SQ/SP testbench.

Change 83566 on 2003/02/11 by hartogs@fl_hartogs2

Modified dump sp_sx.dmp to be more friendly for testbench trackers.

Change 83491 on 2003/02/11 by llefebvr@llefebvr_r400_emu_montreal

adding first of new context arbitration restriction for position export.

Change 83290 on 2003/02/10 by llefebvr@llefebvr_r400_emu_montreal

changing default primitive type to be PRIM_NROMAL instead of RT.

Change 83285 on 2003/02/10 by mmantor@FL_mmantorLT_r400_win

fixed a bug induced by last change that caused a the program to crash when sq_dumps where not enabled.

Change 83166 on 2003/02/10 by mmantor@FL_mmantorLT_r400_win

1. added state data and missing terms to the sc_packer dump.
2. added a dump for the sx_sp interface that delivers attribute data to the interpolators. Reorganized the sq ProcessPixel and Interpolate code to enable these dumps. Still need to make futher modifications to make dump work properly for cylindrical wrap.

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Change 83000 on 2003/02/07 by sallan@sallen_r400_lin_marlboro

ferret: tp_sq add ferret bits, to be unused soon
clean up a few comparitor outputs

Change 82812 on 2003/02/07 by lilefebvr@lilefebvr_r400_emu_montreal

Added a new performance select counter to count the numbers of stalls due uniquely to position export buffer full in the SX.

Change 82646 on 2003/02/06 by lilefebvr@lilefebvr_r400_emu_montreal

now when overflow shifting we do this signed like the HW.

Change 82206 on 2003/02/05 by lilefebvr@lilefebvr_r400_emu_montreal

added channel mask to the SQ->TP interface for ferret interface checking.

Change 82048 on 2003/02/05 by lilefebvr@lilefebvr_r400_emu_montreal

Masking out bits [31:24] on the read of a CF_LOOP register. Now, the emulator will always return 00 for bits [31:24] when reading such a register.

Change 81837 on 2003/02/04 by lilefebvr@lilefebvr_r400_emu_montreal

Trying to fix cyl wrapping bug. Also removed some warnings in sq_alu.h.

Change 81790 on 2003/02/04 by lilefebvr@lilefebvr_r400_emu_montreal

New muladd and dot code as provided by Tom.

Change 81577 on 2003/02/03 by lilefebvr@lilefebvr_r400_emu_montreal

Emulator not behaving correctly when interpolating more than 1 param (in flat shading mode). This is fixed and I am writing directed tests to make sure HW is correct as well.

Change 81430 on 2003/02/03 by lilefebvr@lilefebvr_r400_emu_montreal

New muladd code with exponent prediction.

Change 81246 on 2003/02/01 by sallan@sallen_r400_lin_marlboro

ferret: comparitor work continues
make files now supported (start to phase out cons)
initial tp4 checking
add emu changes for mem id, etc

Change 81131 on 2003/01/31 by lilefebvr@lilefebvre_laptop_r400_emu

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Added loop relative indexing modes to texture fetches.

Change 81096 on 2003/01/31 by lilefebvr@lilefebvre_laptop_r400_emu

The emulator was doing the AND of all channels in order to kill a pixel or not. The right thing to do is the OR as specified by the DX API.

Change 81054 on 2003/01/31 by lilefebvr@lilefebvre_laptop_r400_emu

Enabling GPR writes on the kill instructions (where previously not writting anything).

Change 81015 on 2003/01/31 by lilefebvr@lilefebvre_laptop_r400_emu

fixing a MOVA problem in the emulator when dealing with large primitives.

Change 80459 on 2003/01/29 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing bug in SQ where SQ was reading instruction 0 when it wanted to do a NOP.
Fixing overflow in DOT introduced by previous change (revert).

Change 80400 on 2003/01/29 by lseiler@lseiler_r400_win_marlboro

fixed linux warnings

Change 80277 on 2003/01/29 by lilefebvr@lilefebvre_laptop_r400_emu

Refined the overflow check for the DOT opcodes add.

Change 80103 on 2003/01/28 by lilefebvr@lilefebvre_laptop_r400_emu

changing the order of the processing of the DOT product to match HW. This does not modify the output.

Change 79829 on 2003/01/27 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing the access violation exception.
Also making RealDOT the default setting for the emulator. Emulator should now be 100% HW accurate in the SP.

Change 79344 on 2003/01/24 by mmantor@fl_mmantorxp_r400_win

changed order of new_vector and deallocate_pc in the 1clk transfer. The sc_random test identified a bug where a new_vector and it's deallocate_pc can come in one transfer and the seq scheduled their execution incorrectly.

Change 78935 on 2003/01/23 by lilefebvr@lilefebvr_r400_emu_montreal

Now flushing denorms resulting of the normalization at the output of the interpolators.
Fix for the depth export.

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Change 78889 on 2003/01/23 by lilefebvr@lilefebvr_r400_emu_montreal

Fixing underflow error in the interpolators.

Change 78854 on 2003/01/23 by lilefebvr@lilefebvr_r400_emu_montreal

Was overwriting the booleans at address 0 with the new ones. This is now working fine.

Change 78816 on 2003/01/23 by lilefebvr@lilefebvr_r400_emu_montreal

Forgot to add the CNDE,GT and DOT2ADD in the non-coissuable instructions list.

Change 78672 on 2003/01/22 by lilefebvr@lilefebvr_r400_emu_montreal

fixing the Frac problem with infinities.

Change 78584 on 2003/01/22 by lilefebvr@lilefebvr_r400_emu_montreal

I had a bug in the read register interface for RT fetch memory. It is now resolved.

Change 78324 on 2003/01/21 by lilefebvr@lilefebvr_r400_emu_montreal

fixing a bug with the _CONST opcodes and a memory export bug when exporting more than 1 pixel per block.

Change 77874 on 2003/01/20 by lilefebvr@lilefebvr_r400_emu_montreal

Fix for the scalar fract.

Change 77517 on 2003/01/17 by mmang@mmang_r400_win

Added new hardware accurate sqrt function code.

Change 77429 on 2003/01/17 by lilefebvr@lilefebvr_r400_emu_montreal

Fixing some FRAC precision issues along with MULADD problem with swizzle of SRCC.

Change 77415 on 2003/01/17 by lseiler@lseiler_r400_win_marlboro2

fixes for more warnings

Change 77293 on 2003/01/16 by lseiler@lseiler_r400_win_marlboro2

Changed XOR to != on lines 133, 513, and 515 to eliminate VCC warning message

Change 77210 on 2003/01/16 by lilefebvr@lilefebvr_r400_emu_montreal

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Added a No_Serial bit to the alloc instruction. By seting this the SQ will not wait for texture data to have returned before allowing the alloc to go thru.

Change 76862 on 2003/01/15 by lilefebvr@lilefebvr_r400_emu_montreal

Tried to do the other opcodes as well where the denorm flushes where not done right. These include max,min,max4,trunc,frac and floor.

Change 76782 on 2003/01/15 by lilefebvr@lilefebvr_r400_emu_montreal

new muladd code and dot4 code. No fonctionnal changes cleanup only. Provided by Tom.

Change 75807 on 2003/01/10 by lilefebvr@lilefebvre_laptop_r400_emu

SX performance counters nomenclature change.
New dot product code (activated using RealDot Reg Key)

Change 75716 on 2003/01/10 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing 2D shader bug where the sequencer was exectuting twice the last instruction.

Change 75569 on 2003/01/09 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing exponent underflow in the mul part of the interpolators.

Change 75305 on 2003/01/08 by lilefebvr@lilefebvre_laptop_r400_emu

Now using the new mulAdd function.

Change 75032 on 2003/01/07 by lilefebvr@lilefebvre_laptop_r400_emu

fixing another problem with the cylindrical wrapping.

Change 74951 on 2003/01/07 by lilefebvr@lilefebvre_laptop_r400_emu

adding scalar_CONST opcodes to the emulator. Not tested.

Change 74888 on 2003/01/07 by lilefebvr@lilefebvre_laptop_r400_emu

Found a major bug in the SQ. This should fix the cyl wrapping problem. Needs to be verified with Jeff.

Change 74756 on 2003/01/06 by lilefebvr@lilefebvre_laptop_r400_emu

wasn't performing a denorm flush to zero on the input of the fract. Also doing it for the floor function as well now.

Change 74712 on 2003/01/06 by lseiler@lseiler_r400_win_marlboro2

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eliminated VCC warning messages

Change 74397 on 2003/01/03 by llefebvr@llefebvr_r400_emu

Added Soft RBBM reset feature to the SQ.

Change 74359 on 2003/01/03 by llefebvr@llefebvr_r400_emu

Adding the additionnal restrictions on the MUL_PREV2 opcode.

Change 74297 on 2003/01/03 by sellen@sallen_r400_lin_marlboro

change all occurances of sbfloat<8,23,127> to proper mfloat<8,23,128>

Change 74163 on 2003/01/02 by llefebvr@llefebvr_r400_emu

I removed the input modifiers on previous scalar for the _PREV scalar opcodes per Tom's request.

Change 74144 on 2003/01/02 by llefebvr@llefebvr_r400_emu

There was a conflict between the counters of the SX and those of the SQ. I made the counters of the SX start at 0x30 to remove the conflict.

Change 73968 on 2002/12/31 by mmantor@FL_mmantorLT_r400_win

fixed for the vertex version of bad pipe to work. The sq changes corrected the steering of vertex data to the correct gpr's and the sx solved a hang when odd number of pipes were disabled and the vgt change corrected the pc deallocation generation circuit

Change 73536 on 2002/12/27 by llefebvr@llefebvre_laptop_r400_emu

New DOT code based on the newest muladd function.

Change 73502 on 2002/12/27 by llefebvr@llefebvre_laptop_r400_emu

Changing the XY import to the SP as per Tom Frinsinger's proposal.

Change 71733 on 2002/12/17 by llefebvr@llefebvre_laptop_r400_emu

New MulAdd has been added to the emulator. It is disabled by default. To enable please set NewMulAdd registry key or environment variable to 1.

Change 71681 on 2002/12/17 by llefebvr@llefebvre_laptop_r400_emu

Flushing denorms to 0 before the SET*,CDN* instructions before doing the test.

Change 71542 on 2002/12/16 by llefebvr@llefebvre_laptop_r400_emu

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The SQ wasn't stuffing the vertices correctly in the GPRs when there where bad pipes.

Change 71161 on 2002/12/13 by llefebvr@llefebvre_laptop_r400_emu

Reverting the -ZERO change for the mul only. Now when doing a mul only +ZERO is added as source C.

Change 70456 on 2002/12/11 by llefebvr@llefebvre_laptop_r400_emu

Adding -0 for straight mul's in HW accurate version instead of 0 for better precision.

Change 70403 on 2002/12/11 by llefebvr@llefebvre_laptop_r400_emu

fixing flushing to 0 the denorm BEFORE the NAN check...

Change 70159 on 2002/12/10 by llefebvr@llefebvre_laptop_r400_emu

Fixing overshifting in the interpolators adder.

Change 69667 on 2002/12/09 by llefebvr@llefebvre_laptop_r400_emu

fixing overshifting in the parameter_sub function of the SX.

Change 69637 on 2002/12/09 by llefebvr@llefebvre_laptop_r400_emu

Fixing underflow condition in the interpolators. Also flushing denorms for scalar table functions.

Change 69604 on 2002/12/09 by llefebvr@llefebvre_laptop_r400_emu

Backing out the change to the muladd that broke many regression tests.

Change 69176 on 2002/12/06 by llefebvr@llefebvre_laptop_r400_emu

adding debug info to the parameter subtract.

Change 69158 on 2002/12/06 by llefebvr@llefebvre_laptop_r400_emu

More HW precision fixes for the interpolators. Now matches perfectly HW for regular cases. Still a problem with very large numbers (but they are not used in the regression yet).

Change 68741 on 2002/12/05 by llefebvr@llefebvre_laptop_r400_emu

providing read path on the RBBM for the instruction store.

Change 68507 on 2002/12/04 by llefebvr@llefebvre_laptop_r400_emu

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The emulator was off by 1 bit in the adder. Need to update golds... They should not mismatch by more than 1 LSB.

Change 68349 on 2002/12/04 by llefebvr@llefebvre_laptop_r400_emu

Error in the emulator implementation of TRUNC.

Change 68191 on 2002/12/03 by llefebvr@llefebvre_laptop_r400_emu

new implementation of the DOT product. Currently turned off by default. Need to set RealDOT to 1 in registry file to turn it on. Changed regular mulladd to reflect implementation changes as well.

Change 68039 on 2002/12/03 by llefebvr@llefebvr_r400_linux_marlboro

fixed a NaN propagation problem (the sign wasn't switched correctly)

Change 67086 on 2002/11/26 by llefebvr@llefebvr_r400_linux_marlboro

fixing typo

Change 67080 on 2002/11/26 by llefebvr@llefebvr_r400_linux_marlboro

fixing HW accurate boundary problems with SUB and range checking INF*0

Change 66324 on 2002/11/22 by llefebvr@llefebvre_laptop_r400_emu

Implemented the performance counters in the SQ and SX.

Change 65882 on 2002/11/21 by llefebvr@llefebvre_laptop_r400_emu

Fixing trunc opcode.

Change 65734 on 2002/11/20 by llefebvr@llefebvre_laptop_r400_emu

added performance counters in SQ.

Change 65556 on 2002/11/20 by llefebvr@llefebvre_laptop_r400_emu

Bug in the emulator dealing with large values being trunc or floored. This should fix a bunch of regression tests.

Change 65367 on 2002/11/19 by llefebvr@llefebvre_laptop_r400_emu

This wasn't hung. Just took a lot of time. I reduced the loop count to something more reasonable. Also changed the R400_NAN to FFC00000.

Change 65261 on 2002/11/19 by llefebvr@llefebvre_laptop_r400_emu

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Fixing the FRAC opcode error with negative numbers.

Change 65199 on 2002/11/19 by llefebvr@llefebvre_laptop_r400_emu

There were two problems with this test:

1) The first pass pixel shader wasn't exporting anything thus the SX wasn't sending anything to the RBs hence the hang. You should never have a dummy pixel shader if there are pixel generated.
2) The time allowed to the test was too short. I incremented it to 10000 (was 1000).

I took the opportunity to do some code cleanup as well. Rerun other memory export tests to make sure I did not break anything.

Change 64957 on 2002/11/18 by llefebvr@llefebvre_laptop_r400_emu

Flipping face bit meaning in SQ (I had it backwards)

Change 64913 on 2002/11/18 by llefebvr@llefebvre_laptop_r400_emu

Fixing 2 HW accurate problems dealing with corner cases on RECIP and RECIPSQRT scalar opcodes.

Change 64795 on 2002/11/18 by mkelly@fl_mkelly_r400_win_laptop

* sq_block_model.cpp was not adding the buffer offset for centriods when centers and centriods are sent.
* fixed gold for r400sc_msaa_8_printtypes_01
* add r400sc_sp_sample_cntl_01 to SC regress_e to lock in this test and help minimize future debugging efforts.

Change 64467 on 2002/11/15 by llefebvr@llefebvre_laptop_r400_emu

Fixing cube bug in SP.

Change 64130 on 2002/11/14 by llefebvr@llefebvre_laptop_r400_emu

Fixing RT interpolation problem.

Change 63968 on 2002/11/14 by llefebvr@llefebvre_laptop_r400_emu

fixing FOG export bug in SX.

Change 63642 on 2002/11/13 by llefebvr@llefebvre_laptop_r400_emu

fixing full absolute constant problems in the emulator.

Change 63217 on 2002/11/11 by llefebvr@llefebvre_laptop_r400_emu

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Fixing an error with loops in the SQ that caused it to loop indefinitely. Also adding some more predication tests.

Change 63153 on 2002/11/11 by lilefebvr@lilefebvre_laptop_r400_emu

fixing error that produces black output whenever using HW accurate mode.

Change 63113 on 2002/11/11 by lilefebvr@lilefebvre_laptop_r400_emu

Submitting an example of how to use the SQ generated counters.

Change 62908 on 2002/11/08 by lilefebvr@lilefebvre_laptop_r400_emu

removed the fix to float before the auto-counters. One must now subtract 2**23 before using them.

Change 62859 on 2002/11/08 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed an exponent underflow problem in the HW accurate interpolators.

Change 62797 on 2002/11/08 by lilefebvr@lilefebvre_laptop_r400_emu

Added the write 0 fonctionnality to the PCs (bit 6 vector destination address).

Change 62709 on 2002/11/08 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing the bug introduced by using the scalar address for the vector data in the GPRs.

Change 62558 on 2002/11/07 by askende@andi_crayola_emu

reversed the order in which the vector and scalar results are written in the GPRs.
Scalar first and Vector second.

Change 62485 on 2002/11/07 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed a predication bug where if a pred set was done last it wouldn't count toward the optimisation.

Change 62057 on 2002/11/06 by lilefebvr@lilefebvre_laptop_r400_emu

Swapping the center/centroid sampling to reflect SC change.

Change 61970 on 2002/11/06 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing the interpolators problem. This problem only shows up in tests that change the sampling modes (center/centroid).

Change 61027 on 2002/11/01 by lilefebvr@lilefebvre_laptop_r400_emu

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Flipped GPR_MANAGEMENT fields to match CP.
Removed float to fix on the XY load.
Modified tests that used the feature accordingly.

Change 60854 on 2002/10/31 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing the sending CENTERS only to the interpolators. Previously the emulator would only receive centers if centroids were also sent. Now you can send only the centers.

Change 60720 on 2002/10/31 by lilefebvr@lilefebvre_laptop_r400_emu

added 2 bits of precision to the muladd to match R300 vertex shader precision.

Change 60690 on 2002/10/31 by lilefebvr@lilefebvre_laptop_r400_emu

Problem with the predicate optimization resulting in this hang.

Change 60388 on 2002/10/30 by lilefebvr@lilefebvr_r400_linux_marlboro

fixing HW accurate bug on Linux.

Change 60279 on 2002/10/30 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed the _prev opcodes. Now using the getSignbit instead of isNeg for NAN compatibility.

Change 60038 on 2002/10/29 by lilefebvr@lilefebvre_laptop_r400_emu

Renamed Real time parameter registers to be more explicit.
Implemented alpha to mask in the SX.
Implemented ALPHA_NO_BLEND in the SX.

Change 59911 on 2002/10/29 by mmantor@mmantor_r400_win

sq_block_model.cpp - made separate line counters for loading of sq pixel vector data with a buffer count for the optimized quad transfers when quads from different rows can be sent together.

sx_block_mode.cpp/h seperated the quad fifo's so that even odd pairs are always read together to prevent reordering in the rb's
sc files added bad pipe to the packer and the optimized quad xfers

Change 59731 on 2002/10/28 by lilefebvr@lilefebvre_laptop_r400_emu

Memory overrun in the interpolators that was corrupting the state of the SQ. Also change the pred_clr opcode to write MAX_FLOAT instead of 5000 in the GPRs.

Change 59460 on 2002/10/25 by hartogs@fl_hartogs

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Pipe Disable Change. This change creates a separate set of ROM_SP_disable bits for the vertex shader path and the pixel shader path.

Change 59417 on 2002/10/25 by lilefebvr@lilefebvre_laptop_r400_emu

Implemented the 4 IJ buffer performance optimization in the emulator.

Change 59135 on 2002/10/24 by lilefebvr@lilefebvre_laptop_r400_emu

Using getField instead of getReal for NAN detection in hope of fixing linux HW accurate bug.

Change 59077 on 2002/10/24 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing a problem in the SQ RS management where the call return address wasn't save correctly in some cases. Fixes the advanced_test.

Change 58939 on 2002/10/23 by lilefebvr@lilefebvre_laptop_r400_emu

Underflow of the exponent fixed.

Change 58930 on 2002/10/23 by lilefebvr@lilefebvre_laptop_r400_emu

Fixes in HW accurate stuff.

Change 58812 on 2002/10/23 by lilefebvr@lilefebvre_laptop_r400_emu

Implemented Cube Opcode.
Added CNTX register for reading of the CST store using the RBBM bus for debug purposes.

Change 58676 on 2002/10/22 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing a problem where the predicate bits were not set correctly wich caused problems with predicated texture fetches.

Change 58567 on 2002/10/22 by lilefebvr@lilefebvre_laptop_r400_emu

Pulsing the RB for CACHE flushes events in the VTX shader (and derivative flush events).

Change 58306 on 2002/10/21 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed GradFills and Floor vector instruction.

Change 58287 on 2002/10/21 by lilefebvr@lilefebvre_laptop_r400_emu

Primlib was not passing NANDs correctly for ALU constants.
State changes are no more required for memory exports. I changed primlib to reflect this.

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Change 58052 on 2002/10/18 by lilefebvr@lilefebvre_laptop_r400_emu

ABS modifier now applies to both constants.
Added aperture checks to prevent memory spills for memory exports.

Change 58005 on 2002/10/18 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing r400vgt_vtx_export_very_very_simple_04 memory export test.

Change 58001 on 2002/10/18 by lilefebvr@lilefebvre_laptop_r400_emu

Various fixes to the assembler and the emulator to allow for debug exports.

Change 57713 on 2002/10/17 by lilefebvr@lilefebvre_laptop_r400_emu

added a debug address export destination

Change 57655 on 2002/10/17 by lilefebvr@lilefebvre_laptop_r400_emu

allowing PC allocs in the debugging mode.

Change 57534 on 2002/10/16 by lilefebvr@lilefebvre_laptop_r400_emu

allowing position allocs in debug mode.

Change 57526 on 2002/10/16 by lilefebvr@lilefebvre_laptop_r400_emu

fixing a debug export bug (wasn't letting position thru).

Change 57498 on 2002/10/16 by lilefebvr@lilefebvre_laptop_r400_emu

Added Predicate Exits support for loops in the emulator.

Change 57434 on 2002/10/16 by lilefebvr@lilefebvre_laptop_r400_emu

Not doing the range checking correctly for scalar ops. Was causing an error in 1 vgt test.

Change 57142 on 2002/10/15 by lilefebvr@lilefebvre_laptop_r400_emu

Added the _IEEE compliant opcodes to the scalar pipe. Updated Primlib in consequence.

Change 56880 on 2002/10/14 by lilefebvr@lilefebvre_laptop_r400_emu

Loops, jumps and calls are now using a 13 bit address which allows to jump and call and loop around any control flow addresses (does not requires to be even anymore).

Change 56697 on 2002/10/11 by lilefebvr@lilefebvre_laptop_r400_emu

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Added a SQ_RB pulse interface for memory export synchronization.

Change 56607 on 2002/10/11 by lilefebvr@lilefebvre_laptop_r400_emu

Widened the event interfaces from 4 to 5 bits.

Change 56332 on 2002/10/10 by lilefebvr@lilefebvre_laptop_r400_emu

Missblending the fog in the color. Was shifting and shouldn't.

Change 56280 on 2002/10/10 by lilefebvr@lilefebvre_laptop_r400_emu

HW accuracy problem in interpolators fixed (HOS tests).
Abs is now executed before the negate.

Change 56093 on 2002/10/09 by lilefebvr@lilefebvre_laptop_r400_emu

HW accurate bug fix when multiplying a number by 0 the fn was returning a very small but non-zero number.

Change 56036 on 2002/10/09 by lilefebvr@lilefebvre_laptop_r400_emu

Added the context ID to the SQ->TP interface. The name of the signal is SQ_TP_ctx_id. This is needed for multisampling resolves.

Change 56028 on 2002/10/09 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed Nan*0 exception case in the HW accurate mode of the muladd.

Change 55864 on 2002/10/08 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed a hang in the SQ where if you didn't to a Tfect because of control flow the SQ would wait forever after the TP. This enables the use of the new 2D pixel shader.

Change 55794 on 2002/10/08 by lilefebvr@lilefebvre_laptop_r400_emu

Added range check.
Also the test was clamping the values to infinities where clamped prior to the interface.

Change 55320 on 2002/10/04 by lilefebvr@lilefebvre_laptop_r400_emu

Updated the DOT2ADD opcode to meet new specification.
Checking in the test that reproduces the driver's setup for multiple textured triangles.

Change 55267 on 2002/10/04 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed a bug introduced in vgt400_hos_PNT_01. The arbiter wasn't setting deallocation counts right.

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Change 55126 on 2002/10/03 by lilefebvr@lilefebvre_laptop_r400_emu

multiple fixes for memory exports.

Change 54892 on 2002/10/02 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed some memory export bugs.

Change 54819 on 2002/10/02 by lilefebvr@lilefebvre_laptop_r400_emu

fixing sub scalar.

Change 54748 on 2002/10/02 by lilefebvr@lilefebvre_laptop_r400_emu

Added a clear vertex ready counters in order to clean up the SQ counters on context changes.

Change 54680 on 2002/10/01 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed problems with negative numbers in the muladd HW accurate routine.
Fixed HW accurate problem in the TP where it was not reading from the good vertex buffer index in HW accurate mode. This changes makes the whole mini-regression work in full HW accurate mode.

Change 54604 on 2002/10/01 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing a HW accuracy bug in the interpolators. The mantissa of the A-B and A-C was subshifted by one.

Change 54515 on 2002/10/01 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed HW accuracy problem in mull_add function where the wrong variable was used to shift the mantissa of the result of the mul prior to the add.

Change 54332 on 2002/09/30 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed a bug the driver's compiler found where the SQ wasn't stopping on an EXEC_END in some cases.
Also fixed the arbiter's rules to make sure the new PC allocation scheme was respected.

Change 54149 on 2002/09/27 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed the Sequencer to allow for early allocation of the PC. Allowing memory exports to occur at any point.

Change 54018 on 2002/09/27 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed an error in the control flow machine where the last bit was set too early.

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Change 53879 on 2002/09/26 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed the SETs opcodes. Cleaned up the SX. Added an address buffer for memory exports.

Change 53821 on 2002/09/26 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed the "last" bit warning that was not placed for the right condition.

Change 53670 on 2002/09/25 by lilefebvr@lilefebvre_laptop_r400_emu

The SQ was misreading the SQ_SAMPLING register wich was causing the interpolation of the data with the wrong IJs. Also fixed prmlib to add the 1 parameter only SET and KILL scalar instructions.

Change 53592 on 2002/09/25 by lilefebvr@lilefebvre_laptop_r400_emu

corrected some minor errors in the vertex shader code for multiple state exports.

Change 50793 on 2002/09/11 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing XY reads in the shader pipe.

Change 50772 on 2002/09/11 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing the CP 2D test problem.

Change 50725 on 2002/09/11 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed SQ dump file to output generated parameters as well.

Change 50697 on 2002/09/11 by lilefebvr@lilefebvre_laptop_r400_emu

Fixing the CP 2D bugs.

Change 50488 on 2002/09/10 by lilefebvr@lilefebvre_laptop_r400_emu

Added data dependant masks to the PCs.

Change 50464 on 2002/09/10 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed HW accurate bug in the interpolators that caused inaccuracies whenever the IJs where small.

Fixed the SP_SX interface to make it more HW accurate for the SX-RBRC testbench.
Made the SX more flexible for multiple pixel exports to the same location.

Change 50284 on 2002/09/09 by lilefebvr@lilefebvre_laptop_r400_emu

correction to the sub-normalized number interpretation of the IJs.

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Change 50231 on 2002/09/09 by lilefebvr@lilefebvre_laptop_r400_emu

Corrected bad overflow handling in the HW accurate MULADD.

Change 50050 on 2002/09/06 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed SWAP error that caused mismatches with gold images.
Added count reset capability in the SQ for multipass.
Added better comments for the PARAM_SHADE register field.
Fixed some HW accurate bugs.
Added new MOVA write back to GPRs feature.

Change 49961 on 2002/09/06 by ctaylor@fl_ctaylor_r400_dtwinn_marlboro

Make MPASS_PIX_VEC_PER_PASS 20 bits since SQ auto-inc cannot be any bigger due to FltPt restrictions.

Add PA_SC_CNTL_STATUS.MPASS_OVERFLOW status flag.
Changed name and added new event for controlling MPASS pixel shaders and SQ vertex and pixel counters.

Added SC->CP VizQuery and MP PixShader dumps.
Made window_offset register fields 15 bits instead of 16.
Added SC MP PixShader logic.

Added new signal to SC->SQ interface to prevent incrementation of pixel count for "discarded" MP Pix Shader Pixel Vectors.
Fixed MSAA bug where samples 4-7 were not being set to 0 when MSAA was disabled.
Fixed both EMU and RTL.

Change 49761 on 2002/09/05 by sellen@sallen_r400_lin_marlboro

ferret: add some interfaces for gc comparitors

Change 49690 on 2002/09/05 by lilefebvr@lilefebvre_laptop_r400_emu

Added HW accurate scalar operations but for the SQRT. Tested Log and Exp to some extent. Using Khan 6X Scalar implementation.

Change 49621 on 2002/09/05 by lilefebvr@lilefebvre_laptop_r400_emu

Added GEN_INDEX_VTX field to the SQ_PROGRAM_CNTL register.
Changed name of GEN_INDEX field to GEN_INDEX_PIX.

Change 49556 on 2002/09/04 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed another Linux compilation error

Change 49549 on 2002/09/04 by lilefebvr@lilefebvre_laptop_r400_emu

removed some warnings and a Linux error.

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Change 49540 on 2002/09/04 by llefebvr@llefebvre_laptop_r400_emu

Updated the SX->SQ position ready interface to add one more bit.
Added code for the HW accurate scalar pipe (not used yet).

Change 49374 on 2002/09/03 by mmantor@mmantor_r400_win

added sc output of centers and xy data. Also passed the xy data on the ij bus between the SC and SQ

Change 49353 on 2002/09/03 by llefebvr@llefebvre_laptop_r400_emu

Added Idle0 and Idle1_7 functions for SQ idle status report.

Change 49276 on 2002/09/03 by llefebvr@llefebvre_laptop_r400_emu

Added the RECIP_FF and RECIPSQ_FF scalar opcodes.
Reorganized the opcode enums to match most recent HW.
Added a more detailed description of the EO_RT control register.
Updated prmlib to reflect the opcode changes.

Change 49038 on 2002/08/30 by llefebvr@llefebvre_laptop_r400_emu

Added debug dumps for interpolator inputs and vertex shader exports to parameter caches.

Change 48886 on 2002/08/29 by llefebvr@llefebvre_laptop_r400_emu

Implemented the -MAX_FLOAT changes for the MUL_PREV2, LOG and EXP opcodes.
Implemented the HW accurate version of DOT3, DOT4 and DOT2ADD. All vector opcodes are now HW accurate.

Change 48796 on 2002/08/29 by llefebvr@llefebvre_laptop_r400_emu

Changed to MUL_PREV2 instruction to output -infinity also if SrcC is < 0 (per Andy Gruber's request).

Change 48770 on 2002/08/29 by llefebvr@llefebvre_laptop_r400_emu

Implemented the new "9 bits constant" mode. If set we use the constant address as a real absolute address in the constant store (we do not add the VS_CONST_BASE or PS_CONST_BASE to the address specified in the instruction). Only available for constant A.

Change 48237 on 2002/08/27 by llefebvr@llefebvre_laptop_r400_emu

Added Event filtering in the SQ.

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Added better Idle Status reporting in the SQ that now takes the VGT interface into account (previous version was not taking the staging registers into account just the interface).

Change 48140 on 2002/08/26 by llefebvr@llefebvre_laptop_r400_emu

Fixed 3 bugs in the HW accurate interpolators:
1) Detection of I and J == 0 was wrong
2) Detection of A == 0 was wrong
3) Denormalization of second paramter was done in the inverse direction of what it should have been.
This fixes both prmlib_template_simple_triangle.cpp and milestone_tri.cpp tests

Change 48080 on 2002/08/26 by llefebvr@llefebvre_laptop_r400_emu

There was an indexing problem with the writting of the STs in the GPRs.

Change 47880 on 2002/08/23 by llefebvr@llefebvre_laptop_r400_emu

This is the real fix for bug 299. The other one created nasty side effects in the VGT.

Change 47751 on 2002/08/23 by llefebvr@llefebvre_laptop_r400_emu

There was an error in the SQ that made it wrap around when the SQ_PS/VS_CONST.BASE value was too high. This is fixed.

Change 47592 on 2002/08/22 by llefebvr@llefebvre_laptop_r400_emu

NewVector was not handled correctly when not associated with a pixel vector.

Change 46827 on 2002/08/19 by llefebvr@llefebvre_laptop_r400_emu

For SX quad Fifo reasons we are now forcing memory exports to occur in order.

Change 46728 on 2002/08/19 by llefebvr@llefebvre_laptop_r400_emu

Fixed another MAX GPR bug in the Vertex shader.

Change 46550 on 2002/08/16 by llefebvr@llefebvre_laptop_r400_emu

Added more checks for the Idle SQ function.

Change 46516 on 2002/08/16 by llefebvr@llefebvre_laptop_r400_emu

Fixed the maximum GPR (64) allocation problem. One can now run shaders that use 64 GPRs. The meaning of the VS/PS_NUM_REG has changed it now represents the INDEX OF THE MAXIMUM GPR NUMBER. This change will break the HW.

Change 46315 on 2002/08/15 by llefebvr@llefebvre_laptop_r400_emu

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The SQ->SC dump file was not showing all transactions that occurred. This is now corrected. Also changed the default 2D shader directory (if your SBRANCH and SROOT variables aren't set).

Change 46246 on 2002/08/15 by llefebvr@llefebvre_laptop_r400_emu

Fixed an instruction wrapping problem in the SQ

Change 46140 on 2002/08/14 by llefebvr@llefebvre_laptop_r400_emu

Changed the MASK mnemonics to KILL.
Added DST opcode.
Added MUL_PREV2 opcode.
Reordered the opcodes in prmlib and SP.
Implemented the new KILL and SET SCALAR opcodes, they are now all comparing the ALPHA channel to 0.0f (instead of comapring against the RED channel).

Change 46049 on 2002/08/14 by sellen@sallen_r400_lin_marlboro

ferret: change bool/boolean to boolnumber & test
change pix_mask to TP_SP_pix_mask, etc
more sqspsx tweaks

Change 46036 on 2002/08/14 by llefebvr@llefebvre_laptop_r400_emu

Fixed the clamping problem. Note however that now there is no triangle drawn in the test because the clamping causes the triangle to be degenerate.

Change 45961 on 2002/08/14 by llefebvr@llefebvre_laptop_r400_emu

Now allowing one position vector to be exported in memory export mode to "ping" the PA that everything is done.

Change 45752 on 2002/08/13 by llefebvr@llefebvre_laptop_r400_emu

Signal name change on the TP_SQ interface and backup of the not yet working fully memory export in the SX.

Change 45388 on 2002/08/12 by llefebvr@llefebvre_laptop_r400_emu

Changed the SX allocation strategy to better match the HW in MRT cases. The new allocation scheme puts all MRTs in a single block instead of interleaving the MRTs together like the previous implemetation.

Change 45216 on 2002/08/09 by sellen@sallen_r400_lin_marlboro

tpc_t4 changes for TP_SQ interface

Change 45200 on 2002/08/09 by llefebvr@llefebvre_laptop_r400_emu

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Corrected a bug in MUL_PREV, ADD_PREV and SUB_PREV wich where using incorrect source operands.

Change 45127 on 2002/08/09 by llefebvr@llefebvre_laptop_r400_emu

The SQ now uses the information in state registers to allocate regular PS/VS shaders. The size field is ONLY USED for memory writes.

Change 45015 on 2002/08/08 by llefebvr@llefebvre_laptop_r400_emu

Fixed the fog blend logic in the SP and added SETNEs missing opcode.

Change 44461 on 2002/08/06 by llefebvr@llefebvre_laptop_r400_emu

Fixed a problem in the SQ related to the 2D shader. Now the SQ runs even when the END flag is not associated with any instruction (it adds a NOP to EXEC_ENDs with a count of 0).

Change 44350 on 2002/08/06 by llefebvr@llefebvre_laptop_r400_emu

There was a bug in the SQ that made the last iteration of a loop read invalid loop indexes. This is now fixed.

Change 44321 on 2002/08/05 by askende@askende_r400_linux_emu

added hardware bit accurate implementation for some of the vector instructions

Change 44252 on 2002/08/05 by llefebvr@llefebvre_laptop_r400_emu

New counter based predication scheme.

Change 44232 on 2002/08/05 by llefebvr@llefebvre_laptop_r400_emu

Fixed a parameter generation bug in SQ wich was causing the emulator to crash for test r400su_point_sprite_01.

Change 44054 on 2002/08/02 by llefebvr@llefebvre_laptop_r400_emu

Fixed the emulator to add automatically the generated parameters to the number of interpolated parameters in order not to screw up the parameter cache pointer computation in the VGT when generating parameters.

Change 44026 on 2002/08/02 by llefebvr@llefebvre_laptop_r400_emu

Implemented the new parameter generation scheme. See SQ spec for details.

Change 43826 on 2002/08/01 by askende@askende_r400_linux_emu

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mods related to matching the hardware accuracy Change 43756 on 2002/08/01 by askende@askende_r400_linux_emu new mod on the interpolators to get it to match hardware Change 43742 on 2002/08/01 by sallén@sallen_r400_lin_marlboro ferret: move HW related files from test_lib/tools/ferret to parts_lib/src/test/ferretutils make associated changes add interface changes for tp4_tc testbench make some tweaks for ge testbench Change 43661 on 2002/08/01 by llefebvr@llefebvre_laptop_r400_emu dumping in Hex format Change 43394 on 2002/07/31 by llefebvr@llefebvre_laptop_r400_emu HW accurate interpolators in the SQ. To turn on, set HardwareAccurate environment variable to 1 (or HKEY_LOCAL_MACHINE\SOFTWARE\ATI Technologies\Emulator\Debug\HardwareAccurate to 1 if in windows). Off by default. Change 43234 on 2002/07/30 by llefebvr@llefebvre_laptop_r400_emu The LOD correction bits are now correctly propagated to the TP. Change 43129 on 2002/07/30 by llefebvr@llefebvre_laptop_r400_emu Initialized the staging registers to prevent the NAN error message in the interpolators to be issued without a valid reason. Corrected a typo wich prevented the control flow booleans to be written correctly. Change 42994 on 2002/07/29 by llefebvr@llefebvre_laptop_r400_emu Added the register to disable HW detection of PV/PS. Fixed infinite looping problem when ending a shader on a NOP. Fixed the BW jump problem. Change 41911 on 2002/07/23 by hwise@fl_hwise_r400_win Fixed type casting bug when getting eventId from vgt_sq_verts_data.VGT_SQ_vsisr_data[0] Change 41828 on 2002/07/22 by mmantor@mmantor_r400_win connected event and event_id into the pa_sc interface fixed misc dump files for test bench working added lod_correct values to sc_sq interface Page 53 of 75 Ex. 2048 --- R400 Sequencer Emulator FH --- folder_history	Change 41596 on 2002/07/19 by llefebvr@llefebvre_laptop_r400_emu Corrected the VGT->SQ event interface. Corrected the GFX_COPY_STATE problem. Change 41029 on 2002/07/17 by llefebvr@llefebvre_laptop_r400_emu Placed the hooks for HW accurate interpolation (not used yet)... Change 40887 on 2002/07/16 by llefebvr@llefebvre_laptop_r400_emu Implemented the new bad pipe interface between the VGT and the SQ. The VGT doesn't pad anymore and so the SQ is responsible to "jump" over a bad pipe when filling the reservation stations. Change 40867 on 2002/07/16 by llefebvr@llefebvre_laptop_r400_emu Minor changes to the addressing routines of the SQ in preparation for multipass implemetation. Change 40695 on 2002/07/15 by llefebvr@llefebvre_laptop_r400_emu Fixed the event interface in the SQ where it was using a NULL pointer as a valid mask bit field wich was then crashing the numerical library. This changes should fix both the vertex and the pixel event interface. The primlib_vgt_event_initiator.cpp now works correctly. Change 39443 on 2002/07/10 by hwise@fl_hwise_r400_win CP, SQ, and Primlib Updates for PM4 packet format changes Packet Changes 1) SET_CONSTANT a) Updated CONST_ID field to include "booleans" and "loop" b) Removed ALU and Texture constant grouping and now use DWORD offset in packet c) Removed CONST_WRITE_ENABLE field from ordinal 1 (the ALU and TEX write enables are now set with the LOAD_CONSTANT_CONTEXT packet) d) Replaced CONST_INDEX with CONST_OFFSET because meaning changed (offset refers to dword offset and index used to refer to constant group. Also real-time vs. non-real-time is implied by the queue from which the packet is read rather than being encoded in the index) 2) LOAD_CONSTANT_CONTEXT a) Updated CONST_ID field to include "booleans" and "loop" b) Removed ALU and Texture constant grouping and now use DWORD offset in packet c) Replaced CONST_INDEX with CONST_OFFSET because meaning changed Page 54 of 75 Ex. 2048 --- R400 Sequencer Emulator FH --- folder_history
(offset refers to dword offset and index used to refer to constant group. Also real-time vs. non-real-time is implied by the queue from which the packet is read rather than being encoded in the index) d) Replaced NUM_CONSTANTS ordinal with NUM_DWORDS where number of constants had to be converted into dwords via microcode e) Send CONST_PREFETCH packet to the Microengine rather than the LOAD_CONSTANT_CONTEXT 3) CONST_PREFETCH a) This is a new packet 4) DRAW_INDX a) Removed INDEX offset ordinal from packet (this is now state that must be updated prior to the draw packet) CP Pre-Fetch Parser 1) Added state to hold ALU and Texture write enables used when parsing SET_CONSTANT and LOAD_CONSTANT_CONTEXT packets 2) When parsing LOAD_CONSTANT_CONTEXT packet, send CONST_PREFETCH packet to the Microengine rather than the LOAD_CONSTANT_CONTEXT CP Microengine 1) Updated logic for reporting Microengine IDLE state 2) Added a lot of support logic for multi-context 3) Fixed IP stack popping bug when returning from subroutine SQ Register Write Decode Logic 1) SQ no longer pads 2 dwords between groups of 6 dwords when decoding register addresses Primlib updates for new packet formats 1) Added BOOLEAN_CONSTANT, and LOOP_CONSTANT to enum CONSTANT_TYPE in render_engine.h 2) Updated RENDER_ENGINE member functions to support new PM4 SET_CONSTANT packet format a) RENDER_ENGINE::Load_Alu_Constants() b) RENDER_ENGINE::Load_Texture_Constants() c) RENDER_ENGINE::Open_Set_Constant_Packet() 3) Updated member function RENDER_ENGINE::Load_Draw_Command() to support new 3D_DRAW_INDX_2 and DRAW_INDX PM4 packet formats CP full-chip tests 1) Updated packet creation within tests to match the PM4 spec changes Change 39130 on 2002/07/09 by llefebvr@llefebvre_laptop_r400_emu Fixed scalar trunc, floor and frac opcodes. Page 55 of 75 Ex. 2048 --- R400 Sequencer Emulator FH --- folder_history	Change 38783 on 2002/07/08 by llefebvr@llefebvre_laptop_r400_emu Changed the TP_SQ/SQ_TP interface rs_line name to thread_id to match the HW. Made the corresponding ferret changes. Change 38277 on 2002/07/05 by llefebvr@llefebvre_laptop_r400_emu Implemented correctly lines and points in the SQ. Change 37942 on 2002/07/03 by llefebvr@llefebvre_laptop_r400_emu New interpolation scheme for ST generation for points and lines. Also made the correct changes for flat shading of points and lines. Change 37862 on 2002/07/03 by llefebvr@llefebvre_laptop_r400_emu Changed the interpolation equation from $C+I*(B-C)+J*(A-C)$ to $A+I*(B-A)+J*(C-A)$ to match with the changes that occurred in the SU to do lines and points cleanly. Change 37822 on 2002/07/03 by llefebvr@llefebvre_laptop_r400_emu Added Call stack size checks in the SQ. Change 35516 on 2002/06/21 by llefebvr@llefebvre_laptop_r400_emu Solidified the interpolators in order to make the HW accurate transition easier. Change 35484 on 2002/06/21 by llefebvr@llefebvre_laptop_r400_emu Predication optimization problem fixed in Sq. Change 35105 on 2002/06/20 by llefebvr@llefebvre_laptop_r400_emu corrected the sq tests to use the new primlib include. Change 34940 on 2002/06/19 by kmahler@kmahler_r400_win_devel_views Shader assembler and SQ block changes to support new ALU packing including new absolute modifier in source operand. This also includes syntax checking for new restrictions on export register usage. That is, the destination operands of both the vector and scalar operations must be an export if one is an export. The building of the implicit NOP had to be changed to support this restriction. Syntax checking was added to ensure that only one constant register uses an absolute modifier. The co-issue delimiter must now be " " for older versions not equal to "1.0". Version "1.0" may use either the new delimiter, " ", or the old one, ":",. Change 34483 on 2002/06/17 by llefebvr@llefebvre_laptop_r400_emu Page 56 of 75 Ex. 2048 --- R400 Sequencer Emulator FH --- folder_history

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Implemented scalar exports

Change 34451 on 2002/06/17 by lilefebvr@lilefebvre_laptop_r400_emu

Changed the encoding of the SQ_PROGRAM_CNTL.VS_EXPORT_CNT register. Now a value of 0 means 1 parameter and so forth. Everyone reading the register must now add 1 to have the correct number of parameters. This allows to cover the range 1->16 with only 4 bits.

Change 34419 on 2002/06/17 by lilefebvr@lilefebvre_laptop_r400_emu

Added MOVA opcode to the vector path and MOVA_TRUNC opcode to the scalar path...

Change 34233 on 2002/06/14 by lilefebvr@lilefebvre_laptop_r400_emu

Implemented all the new predicate set instructions (vector and scalar).

Change 34172 on 2002/06/14 by lilefebvr@lilefebvre_laptop_r400_emu

Added/Removed vector and scalar opcodes to comply with the new ALU format as specified by Andy Gruber.

Change 33840 on 2002/06/13 by lilefebvr@lilefebvre_laptop_r400_emu

implemented more scalar opcodes.

Change 33485 on 2002/06/12 by lilefebvr@lilefebvre_laptop_r400_emu

Implemented most of the remaining vector opcodes. Including pixel kills. The pred_set are still not implemented since they are still changing.

Change 33165 on 2002/06/11 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed parameter cache addressing error when exporting more than one color in the VS.

Change 32964 on 2002/06/10 by lilefebvr@lilefebvre_laptop_r400_emu

Fix for r400vgt_hos_cubic_pos_pnt_discrete_01 crashing when vertexTail == 0.

Change 32633 on 2002/06/07 by lilefebvr@lilefebvre_laptop_r400_emu

added an assert for NaNs in the interpolators.

Change 31997 on 2002/06/05 by lilefebvr@lilefebvre_laptop_r400_emu

Added SQ_prefix to allocation_type enum because of a name clash in windows.h

Change 31843 on 2002/06/04 by lilefebvr@lilefebvre_laptop_r400_emu

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Added CONDITIONAL_EXEC_PRED_NO_STALL and CONDITIONAL_EXEC_PRED_NO_STALL_END control flow opcodes to the SQ per Andy Gruber's request.

Change 31717 on 2002/06/04 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed the ABSOLUTE/RELATIVE name clash in the SQ enum by changing the names to ABSOLUTE_ADDR/RELATIVE_ADDR. Fixed some warnings and fixed the SQ_SC_dec_cnt problem where the SQ should have pulsed twice the SC on a new vector but it was only pulsing once.

Change 31527 on 2002/06/03 by lilefebvr@lilefebvre_laptop_r400_emu

Updated the objParser. Added a bypass of the interpolators if all parameters of the primitive are the same.

Change 31254 on 2002/05/31 by askende@andi_crayola_emu_w

fixed a clamping related bug. Numbers greater than 1.0 weren't being clamped when clamping is on.

Change 31115 on 2002/05/31 by lilefebvr@lilefebvre_laptop_r400_emu

Added safety checks in the sq vertex processing

Change 30964 on 2002/05/30 by lilefebvr@lilefebvre_laptop_r400_emu

Implemented the Event interface in the SQ (for vertices).

Change 30942 on 2002/05/30 by lilefebvr@lilefebvre_laptop_r400_emu

updated the VGT->SQ interface (and corresponding blocks) to match HW in name and size and add the Event field.

Change 30570 on 2002/05/29 by lilefebvr@lilefebvre_laptop_r400_emu

fixed a parameter cache deallocation synchronization problem

Change 30311 on 2002/05/28 by lilefebvr@lilefebvre_laptop_r400_emu

removed useless code...

Change 30310 on 2002/05/28 by lilefebvr@lilefebvre_laptop_r400_emu

The CoissuedInstruction flag was not reset properly in sq_alu.cpp. If one mulAdd was done in the shader the scalar path was turned off for the remainder of the shader program. This is now corrected.

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Change 30106 on 2002/05/26 by ctaylor@fl_ctaylor_r400_dttwin_marlboro

Remove FloatParts usage. concerned about speed/memory rqmts.

Change 30068 on 2002/05/24 by ctaylor@fl_ctaylor_r400_dttwin_marlboro

Update SC and SP to use denormalized SE4M20 (exp bias of 6) for IJ data.

Change 29937 on 2002/05/24 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed a event related problem that would have hanged the sequencer in the future (when events are turned on).

Change 29850 on 2002/05/24 by lilefebvr@lilefebvre_laptop_r400_emu

Changed the SQ_PROGRAM_CNTL.VS_EXPORT_MODE register field to use this enumeration instead:

- 0: Position (1 vector).
- 1: Position (2 vectors) Do not use second position export.
- 2: Position (2 vectors) Use point sprite size.
- 3: Position (2 vectors) Use edge flags.
- 4: Position (2 vectors) Use kill flags.
- 5: Position (2 vectors) Use point sprite size and kill flags.
- 6: Position (2 vectors) Use edge flags and kill flags.
- 7: Multipass.

Changed all emulator files (and primlib) to match with the new setting. No impact on the RTL (not using the feature yet).

Change 29678 on 2002/05/23 by lilefebvr@lilefebvre_laptop_r400_emu

Implemented the debug features of the SQ (untested and unused at the moment).

Change 29461 on 2002/05/22 by lilefebvr@lilefebvre_laptop_r400_emu

Corrected the "opcode 15 not supported" error from the primlib_tex test.

Change 29385 on 2002/05/21 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed an arbitration problem in arbiter where two exporting vectors could have the same thread id (causing a hang in the SX).

Change 29075 on 2002/05/20 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed a problem in the vertex input engine of the SQ where the SQ was invalidating the last vertex sent if the continued signal was asserted.

Change 29039 on 2002/05/20 by lilefebvr@lilefebvre_laptop_r400_emu

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Added support for RECIPSQ, LOGs, EXPs. Software implementation only using the C standard funtions.

Change 28843 on 2002/05/17 by lilefebvr@lilefebvre_laptop_r400_emu

Documented all control flow instruction types in sq.blk (and sq.desc).

Change 28778 on 2002/05/17 by askende@andi_crayola_emu_w

fixed the swizzle logic

Change 28375 on 2002/05/16 by lilefebvr@lilefebvre_laptop_r400_emu

Changed the Allocate_size field of the Alloc instruction to match the encoding of the HW.

The encoding is now:
00 -> 1 buffer
01 -> 2 buffers
...
15 -> 16 buffers

Change 28208 on 2002/05/15 by lilefebvr@lilefebvre_laptop_r400_emu

Added the enumeration fields in autoreg for both ALU and Control Flow instructions. Not a fonctionnal change.

Change 28110 on 2002/05/15 by lilefebvr@lilefebvre_laptop_r400_emu

Replaced the reduce precision delat interpolation scheme with full precision interpolation.

Change 27976 on 2002/05/14 by lilefebvr@lilefebvre_laptop_r400_emu

New SQ register map.
Added EXEC_END,CEXEC_END, CPEXEC_END. Removed the END instruction.

Change 27875 on 2002/05/14 by jhoule@jhoule_r400_win_marlboro

Sequencer was calling GetNewTP_SQ_pix_mask instead of GetTP_SQ_pix_mask. TP returns correct masks now (was previously all 1s to be conservative).

Change 27413 on 2002/05/10 by lilefebvr@lilefebvre_laptop_r400_emu

Added texture pipe predication support.

Change 27370 on 2002/05/10 by jhoule@jhoule_r400_win_marlboro

Added pix_mask support (always returns 1111 from TP).

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Added srcSwizzle (TInstrPacked) called from the SQ (arbiter) in order to keep 3 channels SP->TP path.

Adapted tp_dumps to represent all of those.

Change 27266 on 2002/05/09 by vliu@vliu_r400_cnvliu100_win_cvd

Fixed the compile error after remove of the mc block
- everyone, please do a clobber under emu_lib and rebuild
since the ar_* files under ar_code will still be there if not removed

Change 27260 on 2002/05/09 by llefebvr@llefebvre_laptop_r400_emu

Added the pixel masks to the TP->SQ interface. Implemented the JUMP opcode.

Change 27218 on 2002/05/09 by llefebvr@llefebvre_laptop_r400_emu

Added the W field to the SP->TP interface.
Implemented CALL, LOOP and COND opcodes in the control flow machine.

Change 27011 on 2002/05/08 by llefebvr@llefebvre_laptop_r400_emu

Changed the number of bits of the event_id field from 2 to 4 in the sequencer and the corresponding interface per Mike Mantor's request.

Change 26974 on 2002/05/08 by llefebvr@llefebvre_laptop_r400_emu

Added constant relative indexing via address register support and predication support (vector 0 only) in the SP. The pred_set and mova instructions are still not implemented so this is of little use right now.

Change 26955 on 2002/05/08 by llefebvr@llefebvre_laptop_r400_emu

Added relative constant and GPR addressing capabilities to the SQ.

Change 26797 on 2002/05/07 by llefebvr@llefebvre_laptop_r400_emu

Changed the mbfloats in the interfaces for mfloats in order to use IEEE floats instead of doubles in the SP, SX, and RBs. Added the control_flow_store to the SQ.

Change 26750 on 2002/05/07 by llefebvr@llefebvre_laptop_r400_emu

Corrected a problem in the SQ when loading multiple stages in the vertex staging registers.

Change 26569 on 2002/05/06 by llefebvr@llefebvre_laptop_r400_emu

Added readState and WriteState utility functions to the control flow machine in the SQ.

Change 26522 on 2002/05/06 by llefebvr@llefebvre_laptop_r400_emu

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There was a problem in the SQ whenever there where two EXEC instructions in a row. This is now fixed. It fixed the regression and hopefully will fix all other tests using the same template.

Change 26466 on 2002/05/06 by llefebvr@llefebvr_r400_linux_marlboro

fixed linux compilation errors

Change 26365 on 2002/05/03 by llefebvr@llefebvre_laptop_r400_emu

This is the new control flow sequencer. Expect things to be a bit unstable while this major change settles in. I know I broke 1 regression test (r400vgt_index_size_01) but the integration took so long that I decided to check the change in anyways and fix the problem from the TOTT. Sorry for the inconvenience.

Change 26344 on 2002/05/03 by hartogs@fl_hartogs

Added a count the VGT_SQ interface to verify the proper operation of the interface. The added code is #ifdef'ed out and is not a functional change.

Change 24998 on 2002/04/25 by llefebvr@llefebvre_laptop_r400_emu

Small correction in the SQ again because of a bug in the flat shading control code.

Change 24929 on 2002/04/25 by llefebvr@llefebvre_laptop_r400_emu

Fix for the flat shading (there was a shifting problem in the SQ)

Change 23847 on 2002/04/18 by llefebvr@llefebvre_laptop_r400_emu

Fixed a small bug in the SC that made the SQ stall whenever the IJ buffers where filled up.

Change 23698 on 2002/04/17 by llefebvr@llefebvre_laptop_r400_emu

Updated the SQ->TP and TP->SQ interfaces to match the HW in names.

Change 23236 on 2002/04/15 by llefebvr@llefebvre_laptop_r400_emu

Added an error check in the SQ to report an error when trying to deallocate an empty parameter cache.

Change 23080 on 2002/04/12 by rramsey@RRAMSEY_P4_r400_win

Fix bug in interp bb logic
Add commentline dump to sc dump class
Add state dump to sc

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Change SQ input to allow new_vector and pc_dealloc to happen on the same quad from the SC. Laurent L. needs to verify that this change is OK, but it fixes the scissor_rect_04 hang and passes chip regression.

Change 23010 on 2002/04/12 by llefebvr@llefebvre_laptop_r400_emu

Added the #define in vgtout that makes the deallocation of the PC occur on the last primitive instead of having to add another null prim.

Change 22994 on 2002/04/12 by llefebvr@llefebvre_laptop_r400_emu

Added the SQ->SC interface dump and corrected a small bug in the other interface dumps of the SQ,SX that made the dumps slightly out of sync with what was actually sent over the interfaces.

Change 22976 on 2002/04/11 by hartogs@fl_hartogs

* sq_block_model.cpp --> Flipped #if switch to new dealloc mode. Fixed variable name typo.

* interconnect.cpp, interconnect.h --> Added DumpComment() routine to the Interconnect

class to allow comments in the dump file. I used this to print packet boundary comments into the VgtPaClipP dump for easier cross-checking with the other dumps.

* vgt_pa_if.h --> Changed the "dealloc_slot" field from a 'Bool' to an unsigned long.
* vgtout.h, vgtout.cpp --> Coded new dealloc mode.
* vgt_block_model.cpp --> Changed deallocate_slot from bool to ulong.
* vgt_pa_clip_prim.h --> Changed deallocate_slot from bool to ulong.
* For this check-in, the dealloc value can be greater than one, however, the dealloc does NOT occur on the prim using the verts.

Change 22823 on 2002/04/11 by llefebvr@llefebvre_laptop_r400_emu

Added the SC_SQ_provok_vtx to the SC->SQ interface (Randy you must set the correct value if you want flat shading to work).

Added the following interface dumps to the SX:
SX->RB
SX->RB_quads

Added the two ring wrapping modes for the SQ instruction store.

Change 22547 on 2002/04/09 by llefebvr@llefebvre_laptop_r400_emu

Added the capability to dump interfaces to a file to the SQ and SX blocks.

The SQ can now dump the following interfaces:
SQ->SX

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SP->SX
SQ->TP

The SX can now dump the following interface:
SX->PA

Change 22459 on 2002/04/09 by llefebvr@llefebvre_laptop_r400_emu

Added the auto-count to the vertex shaders. The auto-generated counter is loaded in R2.x if SQ_IMPORTS_EXPORTS.GEN_INDEX is set.

Change 22449 on 2002/04/09 by llefebvr@llefebvre_laptop_r400_emu

Added the new PC dealloc scheme (the old one is still there and is the one active for now).

Change 22354 on 2002/04/08 by llefebvr@llefebvre_laptop_r400_emu

Added the cylindrical wrapping logic in the interpolators.

Change 22138 on 2002/04/05 by llefebvr@llefebvre_laptop_r400_emu

Modified the SQ_IMPORTS_EXPROTS.VS_EXPORT_MODE register to reflect the proposed change of the PA review.

Change 22048 on 2002/04/05 by llefebvr@llefebvre_laptop_r400_emu

Added the SQ_CP event interface.
Added support for centers/centroid sampling.
Put provok vertex to 2 if set to last in state (TEMP/CLAY will confirm).

Change 21910 on 2002/04/04 by llefebvr@llefebvre_laptop_r400_emu

Implemented:
1) Real time interpolation.
2) Auto counters for pixel vectors.
3) Flat Shading.
4) Faceness buffers
5) XY buffers
6) Sprite texture coordinate generation

Change 21673 on 2002/04/03 by llefebvr@llefebvre_laptop_r400_emu

Implemented Real time and multi-state management in the SQ (not tested)

Change 21630 on 2002/04/03 by llefebvr@llefebvre_laptop_r400_emu

Added the setContextNumber() functions wherever it was necessary in both the SQ and the SX blocks.

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Change 21539 on 2002/04/03 by lfebevtr@lfebevtr_laptop_r400_emu

Added the event pipelining from the CP (interface back to the CP not done yet).

Change 21480 on 2002/04/02 by rramsey@RRAMSEY_P4_r400_win

Remove OLD_SC stuff from PA and make sure all the sc interface includes are in the new sc directory and being called from there

Change 21275 on 2002/04/01 by lfebevtr@lfebevtr_laptop_r400_emu

Now using the get functions on all context register reads from the SX and the SQ.

Change 20682 on 2002/03/27 by lfebevtr@lfebevtr_laptop_r400_emu

New interfaces between SC,SQ,SP and SX are now implemented and functional.

Change 20483 on 2002/03/26 by jhoule@jhoule_r400_win_marlboro

Implemented new Const/Instr classes.
Integrated with Primlib so that it passes regression.
TFetches don't seem to pass, but VFatches are OK.

Numerous changes:
- Uses const & and plain & for read/write in SQ classes
- Uses accessor methods for set() and get()
- Created new Instr/Const classes, with inheritance (no more unions)
- Created *Packed classes, which are used in the SQ

Change 20198 on 2002/03/22 by lfebevtr@lfebevtr_laptop_r400_emu

Added masking support for the Parameter caches. Added support to write the color buffers out of order to the Export buffer.

Change 19923 on 2002/03/21 by lfebevtr@lfebevtr_laptop_r400_emu

Added pixel kill functionality in the SX. Fixed the SP->SX interface names to match those of the HW.

Change 19654 on 2002/03/19 by lfebevtr@lfebevtr_laptop_r400_emu

removed debug traces

Change 19617 on 2002/03/19 by lfebevtr@lfebevtr_laptop_r400_emu

Corrected a bug in the SX that was causing the last quad of the triangle to be sent twice to the Rb. Also added debug traces in the SX.

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Change 19600 on 2002/03/19 by sallen@sallen_r400_sun_marlboro

ferret shader pipe testbench work

Change 19449 on 2002/03/18 by lfebevtr@lfebevtr_laptop_r400_emu

Added RC_SC to chip.cpp...

Change 19434 on 2002/03/18 by jhoule@jhoule_r400_win_marlboro

Removed hard-coded texture addresses.
Changed alignment multiply from 6 to 8 in SQ.

Change 18970 on 2002/03/14 by lfebevtr@lfebevtr_laptop_r400_emu

Added the new SC_SX interface. Added SX_RB_quad interface. Added RB_SX interface. Modified SX_RB to reflect the HW interface.

Change 18930 on 2002/03/14 by sallen@sallen_r400_sun_marlboro

add export tweak for shader testing

Change 18463 on 2002/03/11 by sallen@sallen_r400_sun_marlboro

timing tweaks for ferret / shader pipe

Change 18443 on 2002/03/11 by lfebevtr@lfebevtr_laptop_r400_emu

Added shader type to the SP->SX data interface to simplify the life of ferret

Change 18345 on 2002/03/11 by lfebevtr@lfebevtr_laptop_r400_emu

Fixed a reservation station management problem in the sequencer related to pixel shader stalls.

Change 18337 on 2002/03/11 by lfebevtr@lfebevtr_laptop_r400_emu

Added export buffers into the SX block. Split the interface from the SQ/SP to the SX into 2 interfaces (data, control). Repaired compilation problems of ferret because of namespace missuses. Removed the VS_EXPORT_MASK register (replaced by a value of 0 in COUNTx registers).

Change 18326 on 2002/03/11 by sallen@sallen_r400_sun_marlboro

ferret shader pipe drive interp data in 512 chunks

Change 18033 on 2002/03/07 by sallen@sallen_r400_sun_marlboro

ferret update for work with shader pipe

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Change 18029 on 2002/03/07 by askende@andi_crayola_emu_w

updated the ALU instruction opcode enumerated type to match the latest shader spec

Change 18028 on 2002/03/07 by askende@andi_crayola_emu_w

updated the opcode enumerated types to match the latest shader spec

Change 17825 on 2002/03/07 by sallen@sallen_r400_sun_marlboro

ferret - new interface routines
- debugging
- SQ_SP drives only pixel data

Change 17818 on 2002/03/07 by lfebevtr@lfebevtr_laptop_r400_emu

Changed the alu instruction packing so that it is consistent with what was done in the texture instruction packing and with the Shader pipe spec (byte 11 = MSB, byte 0 = LSB).

Change 17757 on 2002/03/06 by lfebevtr@lfebevtr_laptop_r400_emu

Added code to support the Control Flow instruction Execute. This code is not activated yet.

Change 17676 on 2002/03/06 by sallen@sallen_r400_sun_marlboro

ferret updates - emulator building .pipe files

Change 17543 on 2002/03/05 by lfebevtr@lfebevtr_laptop_r400_emu

Changed registers writes to be uint32 instead of uint8 for texture state

Change 17503 on 2002/03/05 by lfebevtr@lfebevtr_laptop_r400_emu

Another try to solve the endianness problem

Change 17498 on 2002/03/05 by lfebevtr@lfebevtr_laptop_r400_emu

endian check for unix...

Change 17398 on 2002/03/04 by sallen@sallen_r400_sun_marlboro

re-add ferret interface class needed

Change 17385 on 2002/03/04 by sallen@sallen_r400_sun_marlboro

ferret update for new interface driven pipe model

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Change 17049 on 2002/02/28 by sallen@sallen_r400_sun_marlboro

clean up places ferret was hacked out
add 1st stage of interface based file/pipe in ferret
fix a few make file gotchas
fix a makefile out of order build that prevented a clean build from working

Change 16800 on 2002/02/27 by hartogs@fl_hartogs

Split VGT into its own block.

Change 16798 on 2002/02/27 by sallen@sallen_r400_sun_marlboro

ferret interfaces driven from run.cpp now
change Interface base class to drive ferret data

Change 16466 on 2002/02/25 by jhoule@jhoule_r400_win_marlboro

New 1.30 instruction.
Started integrating common instr/const class shared between TP and PrimLib.

Change 16385 on 2002/02/22 by vliu@vliu_r400_cnviu100_win_cvd

Added support for BUILD_CONFIG environment variable.

Change 16305 on 2002/02/22 by jhoule@jhoule_r400_win_marlboro

Changed TPConst and TPInstr to have:
- pack and unpack instead of writeTo and readFrom
- packing to vector<uint32>&
- DWORD granularity function calls

Change 16176 on 2002/02/21 by jhoule@jhoule_r400_win_marlboro

Changed tp_const and tp_instr to be in a library in cmn_lib/src.
That way, code can be shared between emulator and primlib.

Change 15645 on 2002/02/15 by jhoule@jhoule_r400_win_marlboro

Removed relative paths for numbers.h (includes Charlton Wang's changelist)

Change 15436 on 2002/02/14 by lfebevtr@lfebevtr_laptop_r400_emu

Corrected a small bug in position export.

Change 15382 on 2002/02/13 by lfebevtr@lfebevtr_laptop_r400_emu

Fixed an error in the constant load of the SQ.

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PROTECTIVE ORDER MATERIAL

Change 15305 on 2002/02/12 by lilefebvr@lilefebvre_laptop_r400_emu

Added a count++ in handle_register_fn so that the constant data is written to sq memory...

Change 15275 on 2002/02/12 by lilefebvr@lilefebvre_laptop_r400_emu

Added valid bits between SQ and TP

Change 15268 on 2002/02/12 by lilefebvr@lilefebvre_laptop_r400_emu

Write the vertex data as bytes in the memory

Change 15243 on 2002/02/12 by sallen@sallen_r400_sun_marlboro

add ferret interface class with just one shader pipe

Change 15221 on 2002/02/12 by lilefebvr@lilefebvre_laptop_r400_emu

Updated SQ to take texture write masks into account

Change 15219 on 2002/02/12 by lilefebvr@lilefebvre_laptop_r400_emu

Updated TP->SQ interface to add write masking capabilities

Change 15218 on 2002/02/12 by lilefebvr@lilefebvre_laptop_r400_emu

Corrected an error in texture state management

Change 15217 on 2002/02/12 by lilefebvr@lilefebvre_laptop_r400_emu

The SX now only exports the valid positions to the VGT.

Change 15177 on 2002/02/11 by lilefebvr@lilefebvre_laptop_r400_emu

Interface sync problem on the PA->SQ vertex interface corrected

Change 15149 on 2002/02/11 by ygiang@ygiang_r400_win_marlboro

Added: Ferret hooks to shader Pipe

Change 15072 on 2002/02/11 by lilefebvr@lilefebvre_laptop_r400_emu

Corrected a memory alignment problem that corrupted the state of the arbiter. Also modified simple_triangle to load both the pixel and the vertex shaders.

Change 14992 on 2002/02/08 by lilefebvr@lilefebvre_laptop_r400_emu

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There was an error in the vertex input function of the sequencer it is now fixed and you can load a partial group of vertex.

Change 14979 on 2002/02/08 by lilefebvr@lilefebvre_laptop_r400_emu

Cheking in the last modifications of the SQ SX before trying to integrate the PA block.

Change 14941 on 2002/02/08 by jhoule@jhoule_r400_win_marlboro

Was calling wrong function for filling TPInsr.

Change 14939 on 2002/02/08 by jhoule@jhoule_r400_win_marlboro

iostream.h --> iostream

Change 14791 on 2002/02/06 by lilefebvr@lilefebvre_laptop_r400_emu

Removed all sequencer related texture pipe hardcoded portions of the texture request.

Change 14783 on 2002/02/06 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed the order in which the instructions are written to memory.

Change 14721 on 2002/02/06 by lilefebvr@lilefebvre_laptop_r400_emu

Updated the SQ->SX interface to reflect that the ParameterCaches are now in the SX block.

Change 14627 on 2002/02/05 by lilefebvr@lilefebvre_laptop_r400_emu

Created vertex.sp and pixel.sp. Simplified the shaders to their simplest expression for the 2/22 milestone. Updated the golden images to reflect the change.

Change 14530 on 2002/02/04 by lilefebvr@lilefebvre_laptop_r400_emu

Not seeing the correct data when using IM_LOADS...

Change 14381 on 2002/02/01 by hwise@fl_hwise_r400_win

- 1) Moved RBBM_CNTL and RBBM_SOFT_RESET primary register aperture addresses to match the I/O addresses
- 2) Removed RBBM_*_GO_ASSERT registers
- 3) Added handleRegisterAccess() functions to PA and SQ block classes to catch register read/write access broadcast by RBBM (stub with small decode example)
- 4) More changes to PM4 decode of IM_LOAD and SET_CONSTANT
- 5) Removed a few PM4 type3 packet definitions from prmlib that will not be added to R400 as was previously thought
- 6) Updated prmlib pm4lib.cpp Promo4Lib constructor to

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force the disabling of CP microcode

Change 14318 on 2002/01/31 by lilefebvr@lilefebvre_laptop_r400_emu

Removed all temporary registers from the SQ (but for the _CNT registers wich are needed until we implement CF). Prmlib now needs to send the correct PM4 packets to the CP in order to adjust the SQ registers before sending the first primitive.

Change 14215 on 2002/01/30 by lilefebvr@lilefebvre_laptop_r400_emu

Added multiple triangle support in the interpolators. Also added PC allocation scheme. Also added the SP->Parameter cache dummy interface.

Change 13962 on 2002/01/28 by rbeaudin@rbeaudin_r400_win_marlboro

start of making the transaction engine working

Change 13934 on 2002/01/25 by lilefebvr@lilefebvre_laptop_r400_emu

Added fields to the SQ_SP_Interp interface.

Change 13924 on 2002/01/25 by lilefebvr@lilefebvre_laptop_r400_emu

Added two SQ->SP dummy interfaces to help with HW block level testing.

Change 13871 on 2002/01/25 by jhoule@jhoule_r400_win_marlboro

Now uses standard header <iostream>.

Change 13829 on 2002/01/25 by lilefebvr@lilefebvre_laptop_r400_emu

Added SX_PA interface fixed remaining bugs regarding register integration.

Change 13785 on 2002/01/24 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed the registers but the writes do not go thru because of a prmlib problem?

Change 13725 on 2002/01/24 by hwise@fl_hwise_r400_win

Integrated "play area" register spec into the emulator tree

Change 13669 on 2002/01/23 by lilefebvr@lilefebvre_laptop_r400_emu

Added SX->SQ interface. Updated the Arbiter to take into account co-issue exporting restrictions...

Change 13592 on 2002/01/23 by lilefebvr@lilefebvre_laptop_r400_emu

Added File headers to all SQ and SX files.

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Change 13589 on 2002/01/23 by lilefebvr@lilefebvre_laptop_r400_emu

Added the SX block to the emulator.

Change 13450 on 2002/01/21 by lilefebvr@lilefebvre_laptop_r400_emu

Added the Idle function. Updated sanity. Won't regress because of framebuffer0. Ray is working on it...

Change 13387 on 2002/01/21 by lilefebvr@lilefebvre_laptop_r400_emu

Added Vertex functionalities to the SQ. Remaining interfaces to be added are SQ->SX position and SX->PA position.

Change 13288 on 2002/01/18 by lilefebvr@lilefebvre_laptop_r400_emu

Added Texture pipe fonctionnality to the SQ.

Change 13153 on 2002/01/16 by lilefebvr@lilefebvre_laptop_r400_emu

Added the PA->SQ vertex interface

Change 13149 on 2002/01/16 by lilefebvr@lilefebvre_laptop_r400_emu

Refined the interfaces

Change 12931 on 2002/01/11 by lilefebvr@lilefebvre_laptop_r400_emu

added the GPR allocation for the static mode only for both Vertices and pixel.

Change 12791 on 2002/01/10 by lilefebvr@lilefebvre_laptop_r400_emu

Fixed an interpolation problem when using more than 2 interpolated parameters

Change 12782 on 2002/01/10 by lilefebvr@lilefebvre_laptop_r400_emu

Fix for Jocelyn to work on the TP. This compiles and runs but gives incorect results for now.

Change 12750 on 2002/01/09 by lilefebvr@lilefebvre_laptop_r400_emu

Added TSTATE_MEM_BASE_ADDR register and TSTATE_MEM_WORD_COUNT registers in order to be able to load the texture state from memory. also added a dllexport function in the SQ to to this. Added a texture state store to the SQ.

Change 12674 on 2002/01/09 by lilefebvr@lilefebvre_laptop_r400_emu

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Added a condition in the arbiter that prevents groups of pixel/vertices from passing each other

Change 12633 on 2002/01/09 by llefebvr@llefebvre_laptop_r400_emu

There was a color channel alignment problem in the RB. IT is now fixed but the golden image have to be regenerated yet again. It should be in tones of red and yellow (NOT blue and green).

Change 12617 on 2002/01/08 by askende@andi_crayola_emu_w

rearranged the order of the channels (abgr or wzyx)and completed the mas k capability in the GPR write back logic.

Change 12613 on 2002/01/08 by llefebvr@llefebvre_laptop_r400_emu

channel Write Mask instruction read error in the SQ corrected

Change 12609 on 2002/01/08 by llefebvr@llefebvre_laptop_r400_emu

Changed the channel order to RGBA (R in LSB A in MSB). Emulator wise it means that R is in field 0 and A in field 3.

Change 12605 on 2002/01/08 by llefebvr@llefebvre_laptop_r400_emu

Intermediate changes compiles and runs but doesn't give the right answer

Change 12577 on 2002/01/08 by llefebvr@llefebvre_laptop_r400_emu

Added sources as const variables instead ot regular.

Change 12565 on 2002/01/07 by askende@andi_crayola_emu_w

1.added support for argument selection being a constant

2.added masking capabilities for the write bacI into GPRs

Change 12554 on 2002/01/07 by askende@andi_crayola_emu_w

checking in for backup purposes

Change 12500 on 2002/01/07 by llefebvr@llefebvre_laptop_r400_emu

reduced the size of the SQ_TP interface and added some comments in arbiter.cpp

Change 12441 on 2002/01/04 by askende@andi_crayola_emu_w

Backup do not sync to this changelist it doesn't compile

Change 12388 on 2002/01/04 by llefebvr@llefebvre_laptop_r400_emu

interleaving ALU clauses

Change 12342 on 2002/01/03 by llefebvr@llefebvre_laptop_r400_emu

added ready signal between the PA and the SQ

Change 12328 on 2002/01/03 by llefebvr@llefebvre_laptop_r400_emu

added scalar support in the sequencer. No scalar exports yet.

Change 12177 on 2001/12/21 by llefebvr@llefebvre_laptop_r400_emu

major changes in the SQ block. Now supports clauses. Not fully tested but otherwise operationnal.

Change 12084 on 2001/12/20 by llefebvr@llefebvre_laptop_r400_emu

sbfloats instead of sfloats

Change 12064 on 2001/12/19 by llefebvr@llefebvre_laptop_r400_emu

New interface definition between SQ and TP

Change 12060 on 2001/12/19 by llefebvr@llefebvre_laptop_r400_emu

constants are now working.

Change 12008 on 2001/12/19 by llefebvr@llefebvre_laptop_r400_emu

Changed constants order. Added Constant read support, compiles and runs but doesn't work.

Change 11940 on 2001/12/18 by llefebvr@llefebvre_laptop_r400_emu

repaired a problem with RB valid bits. Added constant support, need to load them using the MC.

Change 11912 on 2001/12/18 by llefebvr@llefebvre_laptop_r400_emu

new ALU instruction format and clamp instructions now in the performance emulator. First golden image can now be generated.

Change 11764 on 2001/12/14 by askende@andi_crayola_emu_w

rearranged the ALU instruction word definition

added the clamping for the ALU results

Change 11581 on 2001/12/12 by rbeaudin@rbeaudin_r400_win_marlboro

changing emu_lib structure